

# THE DENSITY HAJNAL–SZEMERÉDI THEOREM

ABSTRACT. For every fixed integer  $r \geq 3$ , we determine asymptotically, uniformly over  $0 \leq k \leq n/r$ , the maximum number of edges in an  $n$ -vertex graph with at most  $k$  pairwise vertex-disjoint copies of  $K_r$ . Writing  $F_r$  for the explicit maximum of the  $B_{r,i}$ ,  $A_r$ , and  $R_r$  candidate densities, we prove

$$\text{ex}(n, (k+1)K_r) = F_r(k/n)n^2 + o_r(n^2).$$

We also prove the corresponding uniform Erdős–Simonovits stability theorem: every graph whose edge count is within  $o(n^2)$  of this bound differs in  $o(n^2)$  edges from an asymptotically active candidate, whose density gap is  $o(1)$ . At a fixed phase transition, the exact equality classification contains precisely the active extremal families; in particular, the final  $R_r$  phase contains a genuine simplex of generally nonisomorphic extremizers.

The density proof starts from a weighted clique-tiling inequality and the Erdős–Gallai matching theorem. A support-minimal weighted counterexample has an adjacency form that is strictly negative on a codimension-two subspace; interlacing, a sign rule on induced paths, and saturation reduce its support to a clique or a clique blow-up of  $P_3$ . Extreme fractional covers on the latter support have at most two positive levels, after which an explicit positive-semidefinite quadratic form closes the induction. Sampling transfers the weighted theorem to graphons and finite graphs.

For stability, we classify all extremal graphons. Exact first- and second-variation identities handle the open phases. At first-order transitions, a nonsmooth multiplier, an exactly flat four-class measure move, finite uncrossing, and a local Bellman recursion exclude every interior multiplier. All complementary-slackness arguments use near-optimal tilings and therefore do not assume attainment of a graphon tiling optimum. Compactness, cut-to- $L^1$  conversion, and integer rounding then give uniform finite edit stability.

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## 1. INTRODUCTION AND MAIN RESULTS

For graphs  $F, G$ , let  $\nu(F, G)$  be the maximum number of pairwise vertex-disjoint copies of  $F$  in  $G$ , and let

$$\text{ex}(n, (k+1)F)$$

be the maximum number of edges in an  $n$ -vertex graph  $G$  satisfying  $\nu(F, G) \leq k$ . The case  $k = 0$  is the classical Turán problem, while  $F = K_2$  is the matching problem of Erdős and Gallai [5].

The minimum-degree clique-tiling problem is governed by the theorems of Corrádi–Hajnal and Hajnal–Szemerédi [4, 6]. Its density version has several competing extremal constructions. The triangle case was determined by Allen, Böttcher, Hladký, and Piguét [2]; the  $K_4$  case and the general candidate list were obtained by Hou, Hu, Liu, and Zhang [8]. Our first result proves the asymptotic formula for every clique size in one framework.

**Theorem 1.1** (Density Hajnal–Szemerédi theorem). *For every fixed integer  $r \geq 3$ , uniformly over integers  $0 \leq k \leq \lfloor n/r \rfloor$ ,*

$$\text{ex}(n, (k+1)K_r) = F_r(k/n)n^2 + o_r(n^2),$$

where  $F_r$  is defined in [Section 2](#).

The error is uniform in the literal sense that there is a function  $\eta_r(n) \rightarrow 0$  for which the absolute error is at most  $\eta_r(n)n^2$  for every admissible  $k$ . The graphon form, proved first from the weighted theorem, is the following.

**Theorem 1.2** (Graphon density theorem). *For every  $r \geq 2$  and every graphon  $W$ ,*

$$e(W) \leq F_r(\text{til}(K_r, W)). \quad (1)$$

For  $r = 3, 4$  the  $o(n^2)$  error is weaker than the known  $O(n)$  error; the new density content is the upper bound for every  $r \geq 5$ . The exact weighted inequality used in the proof is stated and proved in [Section 3](#).

We next state the equality classification and stability theorem. We use graphon tiling–cover duality, cover attainment, finite-to-graphon transference, and the marked extension of Turán’s theorem; precise formulations and references appear in [Section 7](#).

For  $x \in [0, 1/r]$ , define

$$\mathcal{M}_r(x) := \{W : \text{til}(K_r, W) \leq x, e(W) = F_r(x)\}. \quad (2)$$

The candidate set  $\mathcal{E}_r^*(x)$  is defined in [Definition 2.2](#). The structural core of the paper is the following exact classification.

**Theorem 1.3** (All-parameter extremal graphon classification). *For every fixed  $r \geq 3$  and every  $x \in [0, 1/r]$ ,*

$$\mathcal{M}_r(x) = \mathcal{E}_r^*(x). \quad (3)$$

The compactness consequences of [Theorem 1.3](#) are uniform in the moving parameter and include transition points.

**Theorem 1.4** (Uniform near-active graphon stability). *Fix  $r \geq 3$ . For every  $\varepsilon > 0$  and  $\eta > 0$  there is  $\delta > 0$  such that, for every  $x \in [0, 1/r]$  and every graphon  $W$  satisfying*

$$\text{til}(K_r, W) \leq x, \quad e(W) \geq F_r(x) - \delta,$$

we have

$$\delta_1(W, \mathcal{E}_r(x; \eta)) < \varepsilon.$$

Here  $\mathcal{E}_r(x; \eta)$  is the union of candidate families whose density is within  $\eta$  of  $F_r(x)$ .

**Theorem 1.5** (Uniform moving-parameter finite stability). *Fix  $r \geq 3$ . For every  $\varepsilon > 0$  and  $\eta > 0$  there are  $\delta > 0$  and  $n_0$  such that the following holds. Let  $n, k$  be integers with  $n \geq n_0$  and  $0 \leq k \leq \lfloor n/r \rfloor$ , and let  $G$  be an  $n$ -vertex graph satisfying*

$$\nu(K_r, G) \leq k, \quad e(G) \geq (F_r(k/n) - \delta)n^2.$$

Then there is a candidate type  $\tau$  with

$$f_{r,\tau}(k/n) \geq F_r(k/n) - \eta$$

and an integer candidate graph  $E \in \mathcal{E}_{r,\tau}(n, k)$  such that

$$\text{edit}(G, E) \leq \varepsilon n^2.$$

**Corollary 1.6** (Sequence form). *Let  $G_n$  be a sequence of  $n$ -vertex graphs, let  $k_n$  be integers with  $0 \leq k_n \leq \lfloor n/r \rfloor$ , and put  $x_n = k_n/n$ . Assume  $\nu(K_r, G_n) \leq k_n$  and*

$$e(G_n) = F_r(x_n)n^2 - o(n^2).$$

*Then there are candidate types  $\tau_n$  and corresponding candidate graphs  $E_n$  such that*

$$\frac{\text{edit}(G_n, E_n)}{n^2} + F_r(x_n) - f_{r, \tau_n}(x_n) \longrightarrow 0.$$

**Corollary 1.7** (Phasewise Erdős–Simonovits stability). *Let  $J$  be a compact subinterval of an open phase on which a unique type  $\tau_0$  satisfies  $f_{r, \tau_0} = F_r$ . For every  $\varepsilon > 0$  there are  $\delta > 0, n_0$  such that every  $n$ -vertex graph  $G$ , with integers  $n \geq n_0$  and  $0 \leq k \leq \lfloor n/r \rfloor$ ,  $k/n \in J$ ,  $\nu(K_r, G) \leq k$ , and*

$$e(G) \geq (F_r(k/n) - \delta)n^2,$$

*satisfies*

$$\text{edit}(G, \mathcal{E}_{r, \tau_0}(n, k)) \leq \varepsilon n^2.$$

We organize the proof so that every shared ingredient is established only once. After defining the candidates and their phase diagram, we prove the weighted density theorem and transfer it to graphons and finite graphs. The second half classifies equality graphons by variational certificates, handles all nonsmooth transition points, and then derives uniform graphon and finite stability from the classification. The appendices contain only the three exceptional transition calculations needed in the main proof.

## 2. CANDIDATE FUNCTIONS AND CANDIDATE GRAPHONS

Fix  $r \geq 3$  and  $x \in [0, 1/r]$ . For  $1 \leq i \leq r-1$ , define

$$B_{r,i}(x) = \frac{1}{2} \left( 1 - \frac{(1-ix)^2}{r-i} \right). \quad (4)$$

Define

$$A_r(x) = \frac{r-2}{2(r-1)} + 2x^2, \quad 0 \leq x \leq \frac{1}{2(r-1)}, \quad (5)$$

and, with  $\ell = 1 - rx$ ,

$$R_r(x) = \frac{1}{2} - \ell + \frac{r-1}{2}\ell^2, \quad \frac{r-2}{r(r-1)} \leq x \leq \frac{1}{r}. \quad (6)$$

A curve outside its displayed domain is omitted. Put

$$F_r(x) = \max \left( \{B_{r,i}(x) : 1 \leq i \leq r-1\} \cup \{A_r(x), R_r(x)\} \right). \quad (7)$$

For rank two we use

$$F_2(x) = \max \left\{ x - \frac{x^2}{2}, 2x^2 \right\}.$$

**2.1. The  $B_{r,i}$  graphon.** Let  $s = r - i$ . Partition the probability space into

$$X \sqcup Y_1 \sqcup \cdots \sqcup Y_s$$

with

$$\mu(X) = ix, \quad \mu(Y_j) = \frac{1-ix}{s}.$$

The graphon is 1 on  $X^2$ , on  $X \times Y_j$ , and between distinct  $Y_j, Y_k$ , and is 0 on each  $Y_j^2$ . Denote the resulting graphon by  $W_{B_i}(x)$ , and let  $\mathcal{E}_{r, B_i}(x)$  be its weak-isomorphism class. A fractional cover is  $c = 1/i$  on  $X$  and 0 elsewhere, of total mass  $x$ .

**2.2. The  $A_r$  graphon.** Partition into

$$X \sqcup Z \sqcup Y_2 \sqcup \cdots \sqcup Y_{r-1}$$

with

$$\mu(X) = 2x, \quad \mu(Z) = \frac{1}{r-1} - 2x, \quad \mu(Y_j) = \frac{1}{r-1}.$$

The sets  $X \cup Z, Y_2, \dots, Y_{r-1}$  are the  $r-1$  Turán parts: all distinct parts are completely joined. Inside the first part,  $X$  is a clique,  $Z$  is independent, and  $X$  is anticomplete to  $Z$ . The other  $Y_j$  are independent. Denote this graphon by  $W_A(x)$ , and let  $\mathcal{E}_{r,A}(x)$  be its weak-isomorphism class. The function  $c = 1/2$  on  $X$  and 0 elsewhere is a cover of mass  $x$ .

**2.3. The  $R_r$  family.** Let  $\ell = 1 - rx$ . Partition into

$$X \sqcup \bigcup_{j=1}^{r-1} (Y_j \sqcup Z_j)$$

with

$$\mu(X) = 1 - (r-1)\ell, \quad \mu(Y_j) + \mu(Z_j) = \ell, \quad \sum_{j=1}^{r-1} \mu(Z_j) = \ell.$$

The set  $X$  is a clique. Each  $P_j = Y_j \sqcup Z_j$  is independent, and distinct  $P_j, P_k$  are completely joined. The set  $X$  is complete to each  $Y_j$  and anticomplete to each  $Z_j$ . The family is parameterized by the compact simplex

$$(z_1, \dots, z_{r-1}) \in [0, \ell]^{r-1}, \quad \sum_j z_j = \ell,$$

where  $z_j = \mu(Z_j)$ . Let  $\mathcal{E}_{r,R}(x)$  be the family of all weak-isomorphism classes obtained from this simplex. Put  $c = 1/r$  on  $X \cup \bigcup_j Y_j$  and 0 on  $\bigcup_j Z_j$ .

**Lemma 2.1** (Candidate densities and covers). *For every type  $\tau$  and every parameter in its domain, each  $U \in \mathcal{E}_{r,\tau}(x)$  has a fractional  $K_r$ -cover of mass  $x$  and*

$$e(U) = f_{r,\tau}(x).$$

*Consequently, every active candidate belongs to  $\mathcal{M}_r(x)$ .*

*Proof.* The displayed functions have mass  $x$  and are covers. In the  $B_i$  construction, a clique uses at most one point from each of the  $r-i$  outside independent classes, and hence at least  $i$  points of  $X$ . In the  $A$  construction, a  $K_r$  in the underlying  $(r-1)$ -partite graphon must use two points from the exceptional Turán part; the only adjacent pair inside that part lies in  $X^2$ . In the  $R$  construction no  $K_r$  meets any  $Z_j$ : a point of  $Z_j$  is nonadjacent to  $X$  and to its own independent class, leaving at most one point from each of the other  $r-2$  classes and hence at most  $r-1$  clique vertices in total. Thus every  $K_r$  lies in the region on which the displayed cover has value  $1/r$ .

For the  $B_i$  construction, the complement consists of the  $r-i$  diagonal blocks  $Y_j^2$ , so

$$1 - 2e(U) = \sum_{j=1}^{r-i} \mu(Y_j)^2 = \frac{(1-ix)^2}{r-i}.$$

For the  $A$  construction, start with the balanced complete  $(r-1)$ -partite graphon and add the clique on  $X$ , obtaining

$$e(U) = \frac{r-2}{2(r-1)} + \frac{\mu(X)^2}{2} = A_r(x).$$

For the  $R$  construction, write  $\alpha = \mu(X) = 1 - (r-1)\ell$ . The missing ordered pairs are the diagonal blocks  $P_j^2$  and the two orientations of  $X \times \bigcup_j Z_j$ . Hence

$$1 - 2e(U) = (r-1)\ell^2 + 2\alpha\ell = 2\ell - (r-1)\ell^2,$$

which is equivalent to  $e(U) = R_r(x)$ . If the type is active, then  $e(U) = F_r(x)$  and the cover gives  $\text{til}(K_r, U) \leq x$ , so  $U \in \mathcal{M}_r(x)$ .  $\square$

**2.4. Active and near-active types.** Let

$$\mathfrak{T}_r = \{B_1, \dots, B_{r-1}, A, R\}.$$

Write  $f_{r,B_i} = B_{r,i}$ ,  $f_{r,A} = A_r$ , and  $f_{r,R} = R_r$ , with value  $-\infty$  outside the appropriate domain.

**Definition 2.2.** The active type set and the  $\eta$ -active type set are

$$\mathcal{A}_r(x) = \{\tau \in \mathfrak{T}_r : f_{r,\tau}(x) = F_r(x)\},$$

$$\mathcal{A}_r(x; \eta) = \{\tau \in \mathfrak{T}_r : f_{r,\tau}(x) \geq F_r(x) - \eta\}.$$

The active and near-active graphon families are

$$\mathcal{E}_r^*(x) = \bigcup_{\tau \in \mathcal{A}_r(x)} \mathcal{E}_{r,\tau}(x), \quad \mathcal{E}_r(x; \eta) = \bigcup_{\tau \in \mathcal{A}_r(x; \eta)} \mathcal{E}_{r,\tau}(x).$$

Every candidate graphon is a 0/1 step graphon with at most  $2r - 1$  blocks.

**Lemma 2.3** (Phase diagram). *The function  $F_r$  is continuous and strictly increasing on  $[0, 1/r]$ . Apart from the common Turán degeneration at  $x = 0$ , its active types occur in the order*

$$B_1, \quad A, \quad B_2, \quad B_3, \dots, \quad B_{r-1}, \quad R.$$

The nonzero  $B_1 = A$  crossing is

$$x = \frac{2}{4r - 3},$$

the  $A = B_2$  crossing is tangential and occurs at

$$x = \frac{1}{2(r - 1)},$$

and, for  $2 \leq i \leq r - 2$ , the effective  $B_i = B_{i+1}$  crossing is the unique solution of

$$\frac{1 - ix}{\sqrt{r - i}} = \frac{1 - (i + 1)x}{\sqrt{r - i - 1}}. \quad (8)$$

Finally,  $B_{r-1} = R_r$  at the unique effective root whose zero-set parameter is

$$\ell_* = \frac{r^2 - r + 1 - r\sqrt{r^2 - 3r + 3}}{(r - 1)(r^2 + r - 1)}. \quad (9)$$

There are no other active intersections.

*Proof.* Write

$$d_i(x) = 1 - 2B_{r,i}(x) = \frac{(1 - ix)^2}{r - i}, \quad d_A(x) = \frac{1}{r - 1} - 4x^2.$$

Since  $x \leq 1/r$ , all square roots below are positive. Put  $s = r - i$  and  $\ell = 1 - rx$ . Then

$$\sqrt{d_i(x)} = \frac{1 - ix}{\sqrt{r - i}} = \frac{\ell}{\sqrt{s}} + x\sqrt{s}.$$

The equality of the terms indexed by  $s$  and  $s - 1$  is equivalent to

$$x = \frac{\ell}{\sqrt{s(s - 1)}} \iff x = \frac{1}{r + \sqrt{s(s - 1)}}.$$

As  $s$  decreases from  $r - 1$  to 1, these crossing points strictly increase. Moreover the sequence  $\ell/\sqrt{s} + x\sqrt{s}$  is unimodal in  $s$ : its consecutive difference changes sign exactly at the displayed threshold. Hence the lower envelope visits every  $B_i$  in increasing order of  $i$ , and no nonconsecutive  $B$ -intersection is active.

The assertions involving  $A$  follow from

$$d_1 - d_A = \frac{x((4r-3)x-2)}{r-1}, \quad d_2 - d_A = \frac{(2(r-1)x-1)^2}{(r-2)(r-1)}.$$

The first identity gives the nonzero  $B_1 = A$  crossing. The second shows that  $A$  dominates  $B_2$  up to their tangential meeting at the endpoint of the  $A$ -domain. For  $r = 3$  there is no later  $B$ -branch. For  $r \geq 4$ , the next  $B$ -crossing occurs at  $1/[r + \sqrt{(r-2)(r-3)}] > 1/[2(r-1)]$ , so no later  $B_i$  intervenes while  $A$  is defined.

For the  $R$  comparison, the deficiencies of  $B_{r,r-1}$  and  $R_r$  are

$$d_B = \left( \frac{1 + (r-1)\ell}{r} \right)^2, \quad d_R = 2\ell - (r-1)\ell^2.$$

Their equality is

$$(r-1)(r^2 + r - 1)\ell^2 - 2(r^2 - r + 1)\ell + 1 = 0.$$

The smaller root is (9). The larger root is

$$\ell_+ = \frac{r^2 - r + 1 + r\sqrt{r^2 - 3r + 3}}{(r-1)(r^2 + r - 1)}.$$

It lies strictly outside the  $R$ -domain. Indeed,

$$r^2(r^2 - 3r + 3) - 4(r-1)^2 = (r-2)^2(r^2 + r - 1) > 0,$$

so  $r\sqrt{r^2 - 3r + 3} > 2(r-1)$  and hence  $\ell_+ > 1/(r-1)$ . The smaller root lies inside the domain, since

$$(d_B - d_R) \left( \frac{1}{r-1} \right) = -\frac{(r-2)^2}{r^2(r-1)} < 0 \quad \text{and} \quad (d_B - d_R)(0) = \frac{1}{r^2} > 0.$$

At the last adjacent  $B$ -crossing, where  $\ell = \sqrt{2}/(r + \sqrt{2})$ , one has

$$d_B - d_R = -\frac{2r(\sqrt{2}-1) + 3 - 2\sqrt{2}}{r^2 + 2\sqrt{2}r + 2} < 0,$$

whereas  $d_B - d_R = 1/r^2 > 0$  at  $\ell = 0$ . Thus the unique effective  $B_{r,r-1} = R_r$  crossing is the smaller root, it occurs after  $B_{r,r-1}$  has become the active  $B$ -branch, and  $R_r$  is active precisely for smaller  $\ell$ .

On the interior of every nondegenerate active interval, the active branch has positive derivative. At  $x = 0$ , the simultaneously active  $B_{r,1}$  branch satisfies  $B'_{r,1}(0) = 1/(r-1) > 0$ . Hence  $F_r$  is strictly increasing on  $[0, 1/r]$ . The displayed identities also give continuity at all restricted-domain endpoints.  $\square$

**2.5. Integer candidates.** For  $x = k/n$ , the notation  $\mathcal{E}_{r,\tau}(n, k)$  means the following integer rounding of the corresponding graphon template.

- For  $B_i$ , take a clique  $X$  of size  $ik$  and split the remaining vertices into  $r-i$  independent classes whose sizes differ by at most one; join all pairs prescribed by the  $B_i$  template.
- For  $A$ , take a balanced complete  $(r-1)$ -partite graph, choose a part containing at least  $2k$  vertices, choose  $X$  of size  $2k$  in that part, and add all edges inside  $X$ .
- For  $R$ , put  $L = n - rk$ . Take a clique  $X$  of size  $n - (r-1)L$ , choose integers  $z_1, \dots, z_{r-1} \in [0, L]$  with  $\sum_j z_j = L$ , put  $|Z_j| = z_j$  and  $|Y_j| = L - z_j$ , and impose the  $R$  template.

Each construction has a fractional  $K_r$ -cover of total weight  $k$ . Rounding a graphon block vector to these integer data moves only  $O_r(1)$  vertices between blocks, apart from the unavoidable balanced rounding of equal parts, and therefore changes only  $O_r(n)$  edges.

**Proposition 2.4** (Finite lower bound). *For every fixed  $r \geq 3$ , uniformly over integers  $0 \leq k \leq \lfloor n/r \rfloor$ ,*

$$\text{ex}(n, (k+1)K_r) \geq F_r(k/n)n^2 - O_r(n).$$

*Proof.* Choose a type attaining  $F_r(k/n)$  and take an integer candidate of that type. The displayed fractional cover has total weight  $k$ , so every  $K_r$ -tiling has size at most  $k$ . The graphon density calculation in Lemma 2.1, together with the  $O_r(1)$  rounding of the block sizes, gives  $F_r(k/n)n^2 - O_r(n)$  edges. For the  $R$  type, any integer vector  $(z_1, \dots, z_{r-1})$  with  $0 \leq z_j \leq n - rk$  and  $\sum_j z_j = n - rk$  gives the same main term.  $\square$

### 3. THE WEIGHTED DENSITY THEOREM

This section proves the finite weighted inequality that drives both the density theorem and the later equality analysis. A *weighted graph* is a pair  $(H, p)$ , where  $H$  is a finite simple graph and  $p = (p_v)_{v \in V(H)}$  is a probability vector with every  $p_v > 0$ . Put

$$e(H, p) = \sum_{uv \in E(H)} p_u p_v = \frac{1}{2} p^\top A(H) p.$$

Let  $\mathcal{K}_r(H)$  be the set of copies of  $K_r$  in  $H$ .

A weighted fractional  $K_r$ -tiling is a vector  $t = (t_K)_{K \in \mathcal{K}_r(H)}$  satisfying

$$t_K \geq 0, \quad \sum_{K \ni v} t_K \leq p_v \quad (v \in V(H)).$$

Its maximum size  $\sum_K t_K$  is denoted by  $\text{til}_r(H, p)$ . A fractional  $K_r$ -cover is a vector  $c \in \mathbb{R}_{\geq 0}^{V(H)}$  such that

$$\sum_{v \in K} c_v \geq 1 \quad (K \in \mathcal{K}_r(H)).$$

Its mass is  $\langle p, c \rangle = \sum_v p_v c_v$ , and the minimum cover mass is  $\text{fcov}_r(H, p)$ . Finite-dimensional linear-programming duality gives

$$\text{til}_r(H, p) = \text{fcov}_r(H, p). \tag{10}$$

The constant vector  $1/r$  is a cover, so these quantities belong to  $[0, 1/r]$ .

**Theorem 3.1** (Weighted density theorem). *Let  $r \geq 2$ , let  $(H, p)$  be a weighted graph, and put  $x = \text{fcov}_r(H, p)$ . Then*

$$e(H, p) \leq F_r(x).$$

*Consequently, if  $c$  is any fractional  $K_r$ -cover with  $\langle p, c \rangle \leq 1/r$ , then*

$$e(H, p) \leq F_r(\langle p, c \rangle).$$

#### 3.1. The Erdős–Gallai base and the peeling recursion.

**Proposition 3.2** (Weighted Erdős–Gallai base). *Theorem 3.1 holds for  $r = 2$ .*

*Proof.* Put  $x = \text{til}_2(H, p) = \text{fcov}_2(H, p)$ . First suppose that all masses are rational. Fix a common denominator  $N_0$ , let  $N = tN_0$ , and blow up every vertex  $v$  to an independent class of size  $Np_v$ , joining two classes completely precisely when their support vertices are adjacent. Call the resulting graph  $G_N$ . Then

$$\frac{e(G_N)}{N^2} = e(H, p).$$

Every integral matching in  $G_N$ , divided by  $N$  and grouped by support edges, is a weighted fractional matching in  $(H, p)$ . Thus, with  $m_N = \nu(K_2, G_N)$ ,

$$\frac{m_N}{N} \leq x.$$



The Erdős–Gallai matching theorem [5] gives

$$e(G_N) \leq \max \left\{ \binom{2m_N + 1}{2}, m_N(N - m_N) + \binom{m_N}{2} \right\}.$$

After division by  $N^2$  and passage to a subsequence, write  $m_N/N \rightarrow y \leq x$ . Letting  $t \rightarrow \infty$  yields

$$e(H, p) \leq \max \left\{ 2y^2, y - \frac{y^2}{2} \right\} \leq F_2(x),$$

because both branches are increasing on  $[0, 1/2]$ . Positive rational probability vectors are dense. After truncating covers to  $[0, 1]^{V(H)}$ , the feasible cover polytope is compact and independent of  $p$ , and

$$|\text{fcov}_2(H, p) - \text{fcov}_2(H, p')| \leq \|p - p'\|_1.$$

Together with the continuity of  $e(H, p)$ , this proves the general case.  $\square$

For a function  $f : [0, 1/(r-1)] \rightarrow \mathbb{R}$ , define the peeling operator

$$(\mathcal{T}_r f)(x) = \sup_{\substack{(r-1)x \leq q \leq 1 \\ q > 0}} \left\{ q(1-q) + q^2 f(x/q) \right\}. \quad (11)$$

At  $x = 0$  one may also allow  $q = 0$ , with value zero.

**Lemma 3.3** (Clique-size recursion). *For every  $r \geq 3$  and  $x \in [0, 1/r]$ ,*

$$\mathcal{T}_r F_{r-1}(x) \leq F_r(x).$$

For  $1 \leq i \leq r-2$ , write

$$q_i^* = \frac{r-i-1+ix}{r-i}.$$

At the first-order transitions listed below, equality is rigid as follows. If  $r \geq 4$  and  $B_{r,1}(x) = A_r(x)$ , equality in

$$q(1-q) + q^2 F_{r-1}(x/q) \leq F_r(x)$$

can hold only for

$$q = q_1^* = \frac{r-2+x}{r-1} \quad \text{or} \quad q = \frac{r-2}{r-1}.$$

If  $2 \leq i \leq r-3$  and  $B_{r,i}(x) = B_{r,i+1}(x)$ , equality can hold only for  $q = q_i^*$  or  $q = q_{i+1}^*$ .

*Proof.* We treat the branches of  $F_{r-1}$  separately. For  $1 \leq i \leq r-2$ ,

$$\begin{aligned} & q(1-q) + q^2 B_{r-1,i}(x/q) \\ &= q - \frac{q^2}{2} - \frac{(q-ix)^2}{2(r-1-i)}. \end{aligned} \quad (12)$$

This is concave in  $q$ , with its unique maximum at  $q = q_i^*$ . The identities

$$q_i^* - (r-1)x = \frac{(r-i-1)(1-rx)}{r-i} \geq 0, \quad 1 - q_i^* = \frac{1-ix}{r-i} \geq 0$$

show that this point is feasible, and substitution gives

$$\sup_q \{q(1-q) + q^2 B_{r-1,i}(x/q)\} = B_{r,i}(x). \quad (13)$$

At the lower endpoint  $q = (r-1)x$ , a complete normalized remainder gives

$$q(1-q) + \frac{q^2}{2} = B_{r,r-1}(x). \quad (14)$$

For  $r = 3$ , the remaining branch of  $F_2$  is  $2y^2$ , and its transform is

$$q(1 - q) + 2x^2 = q - q^2 + 2x^2.$$

If  $x \leq 1/4$ , the maximum over  $q \geq 2x$  occurs at  $q = 1/2$  and equals  $A_3(x)$ . If  $1/4 \leq x \leq 1/3$ , it occurs at  $q = 2x$  and equals  $B_{3,2}(x)$ .

Now let  $r \geq 4$ . On the  $A_{r-1}$  branch the domain requires  $q \geq 2(r-2)x$ , and

$$q(1 - q) + q^2 A_{r-1}(x/q) = q - \frac{r-1}{2(r-2)} q^2 + 2x^2. \quad (15)$$

Its unconstrained maximum is at  $q = (r-2)/(r-1)$ . This point is feasible when  $x \leq 1/[2(r-1)]$ , and the value is  $A_r(x)$ . When

$$\frac{1}{2(r-1)} < x \leq \frac{1}{2(r-2)},$$

the maximum is at  $q = 2(r-2)x$ , where

$$A_{r-1}\left(\frac{1}{2(r-2)}\right) = B_{r-1,2}\left(\frac{1}{2(r-2)}\right),$$

so it is already controlled by (13). For larger  $x$  the branch is absent.

For  $r = 4$ , the transformed  $R_3$  branch has domain  $3x \leq q \leq \min\{6x, 1\}$  and equals

$$\Psi_x(q) = (1 - 3x)q - \frac{1}{2}q^2 + 9x^2.$$

If  $0 < x \leq 1/9$ , its maximum occurs at  $q = 6x$ . Since

$$R_3(1/6) = \frac{1}{4} < \frac{47}{144} = B_{3,1}(1/6),$$

this value is at most the transformed  $B_{3,1}$  maximum  $B_{4,1}(x)$ . If  $1/9 \leq x \leq 1/6$ , the stationary point  $q = 1 - 3x$  is feasible and

$$\Psi_x(1 - 3x) = \frac{1}{2} - 3x + \frac{27}{2}x^2,$$

while

$$A_4(x) - \Psi_x(1 - 3x) = -\frac{1}{6} + 3x - \frac{23}{2}x^2.$$

The latter is concave and has positive endpoint values  $2/81$  and  $1/72$ . If  $1/6 \leq x \leq 1/4$ , the maximum occurs at  $q = 3x$  and equals  $B_{4,3}(x)$ .

Finally let  $r \geq 5$ . The transform of  $R_{r-1}$  is a quadratic in  $q$  whose quadratic coefficient is  $(r-5)/2$ . It is affine for  $r = 5$  and convex for  $r \geq 6$ , hence its maximum occurs at a domain endpoint. The lower endpoint  $q = (r-1)x$  is covered by (14). If the upper endpoint comes from the left endpoint

$$y_0 = \frac{r-3}{(r-1)(r-2)}$$

of the  $R_{r-1}$  domain, then

$$B_{r-1,r-2}(y_0) - R_{r-1}(y_0) = \frac{(r-3)^2}{2(r-2)(r-1)^2} > 0,$$

so (13) controls it. If the upper endpoint is  $q = 1$ , then  $x \geq y_0$  and

$$B_{r,r-1}(x) - R_{r-1}(x) = \frac{(1 - (r-1)x)((r-1)^2x - r + 3)}{2} \geq 0.$$

This completes the inequality. We record why the asserted equality sets follow from the same calculation. Each expression in (12) is strictly concave and has the unique maximizer  $q_i^*$ . The expression in (15) is also strictly concave and, when it yields  $A_r(x)$ , has the unique maximizer

$(r-2)/(r-1)$ . At an effective crossing in the statement, [Lemma 2.3](#) says that precisely the two indicated branches equal  $F_r(x)$ ; all other  $B$ - and  $A$ -branches are strictly smaller. There is one additional boundary issue for the transformed  $A_{r-1}$  branch. When its stationary point is infeasible, its maximum is attained at

$$q_A = 2(r-2)x$$

and equals the value of the transformed  $B_{r-1,2}$  branch there. The latter branch has its unique maximum at

$$q_2^* = \frac{r-3+2x}{r-2}, \quad q_A = q_2^* \iff x = \frac{1}{2(r-1)}.$$

Thus at the first-order transition  $B_{r,2} = B_{r,3}$  this boundary value is strictly below  $B_{r,2}(x) = F_r(x)$ ; at every later transition it is at most the already nonactive value  $B_{r,2}(x) < F_r(x)$ .

For  $r \geq 5$ , at the lower endpoint  $q = (r-1)x$  the transformed  $R_{r-1}$  branch equals  $B_{r,r-1}(x)$ , which is strictly below  $F_r(x)$  at the transitions in the statement by [Lemma 2.3](#). At either upper endpoint, strictness follows from the endpoint comparisons above together with the same phase diagram. When  $r = 4$ , the only crossing in the statement is  $B_{4,1} = A_4$  at  $x = 2/13$ ; the interior maximum of the transformed  $R_3$  branch is below their common value by  $23/1014$ . Thus equality occurs only at the displayed values of  $q$ .  $\square$

**Lemma 3.4** (Universal zero peeling). *Let  $r \geq 3$  and assume [Theorem 3.1](#) for clique size  $r-1$ . Suppose  $(H, p)$  has a fractional  $K_r$ -cover  $c$  of mass  $x \in [0, 1/r]$  and has a universal vertex  $b$  with  $c_b = 0$ . Then  $e(H, p) \leq F_r(x)$ .*

*Proof.* Put  $q = 1 - p_b$ . The restriction of  $c$  to  $H - b$  covers every  $K_{r-1}$ , because adjoining  $b$  produces a  $K_r$ . If  $q = 0$ , then  $x = e(H, p) = 0$ . Suppose  $q > 0$ . If  $q < (r-1)x$ , then

$$e(H, p) \leq q(1-q) + \frac{q^2}{2} \leq B_{r,r-1}(x),$$

because  $q - q^2/2$  is increasing on  $[0, 1]$ . If  $q \geq (r-1)x$ , normalize the masses on  $H - b$  by  $q$ . The restricted cover has mass  $x/q \leq 1/(r-1)$ , and the induction hypothesis together with [Lemma 3.3](#) gives

$$e(H, p) \leq q(1-q) + q^2 F_{r-1}(x/q) \leq F_r(x).$$

$\square$

#### 4. A SUPPORT-MINIMAL WEIGHTED COUNTEREXAMPLE

Fix  $r \geq 3$  and assume inductively that [Theorem 3.1](#) holds for clique size  $r-1$ . Suppose that it fails for  $r$ . Since  $F_r(1/r) = 1/2$ , a counterexample exists at some  $\bar{x} < 1/r$ .

Call  $(H, p, c_0)$  a counterexample at  $\bar{x}$  if  $c_0$  is a fractional  $K_r$ -cover with  $\langle p, c_0 \rangle \leq \bar{x}$  and  $e(H, p) > F_r(\bar{x})$ . Choose, among such counterexamples, one with the minimum support order  $v_0$ . On a fixed labeled  $v_0$ -vertex set, maximize  $e(H, p)$  over all triples  $(H, p, c_0)$  in which  $p$  lies in the closed probability simplex and  $c_0 \in [0, 1]^{V(H)}$  is a fractional  $K_r$ -cover with  $\langle p, c_0 \rangle \leq \bar{x}$ . There are finitely many support graphs and each feasible set is compact, so a maximum exists. Its value is greater than  $F_r(\bar{x})$ , and a maximizing mass vector has no zero coordinate: otherwise deleting a zero-mass atom would give a smaller counterexample.

Keep  $(H, p)$  fixed and choose an extreme minimum solution  $c$  of

$$\mathcal{C}_r(H) = \{z \in \mathbb{R}_{\geq 0}^{V(H)} : z(K) \geq 1 \text{ for every } K \in \mathcal{K}_r(H)\}.$$

Such a solution exists. An optimal coordinate cannot exceed one, since truncation to one preserves every constraint and lowers the objective. Thus the optimal face is a nonempty compact polytope in the cube  $[0, 1]^{V(H)}$  and has an extreme point that is also extreme in  $\mathcal{C}_r(H)$ . Put

$$x = \langle p, c \rangle = \text{fcov}_r(H, p) \leq \bar{x}.$$

By the monotonicity of  $F_r$ ,

$$e(H, p) > F_r(\bar{x}) \geq F_r(x). \quad (16)$$

The support has at least three atoms; otherwise  $e(H, p) \leq 1/4 \leq F_r(0) \leq F_r(x)$ .

**Lemma 4.1** (Positive cover coordinates are tight). *Every atom  $v$  with  $c_v > 0$  belongs to a  $K_r$  satisfying  $c(K) = 1$ .*

*Proof.* If every  $K_r$  containing  $v$  had cover sum strictly greater than one, then finiteness would allow us to decrease  $c_v$  slightly while preserving all cover constraints. Since  $p_v > 0$ , this would lower the cover mass.  $\square$

The support  $H$  is saturated with respect to  $c$ : adding any nonedge makes  $c$  cease to be a cover. Indeed, otherwise that edge could be added while preserving feasibility and strictly increasing  $e(H, p)$ .

**Lemma 4.2** (Saturation certificate). *For every nonedge  $uv$  of  $H$ , there is a copy  $Q \cong K_{r-2}$  in  $N_H(u) \cap N_H(v)$  such that*

$$c_u + c_v + c(Q) < 1.$$

*Proof.* Every new  $K_r$  created by adding  $uv$  has the form  $Q \cup \{u, v\}$  with  $Q \cong K_{r-2}$  in the common neighborhood. Since  $c$  is not a cover after the addition, one of these new cliques violates the cover inequality.  $\square$

**4.1. Strict negativity and the co-chain reduction.** Let  $M = A(H)$ .

**Lemma 4.3.** *For every nonzero  $z \in \mathbb{R}^{V(H)}$  satisfying*

$$\langle z, \mathbf{1} \rangle = 0, \quad \langle z, c \rangle = 0,$$

*one has  $z^\top M z < 0$ .*

*Proof.* For sufficiently small positive and negative  $t$ , the vector  $p + tz$  is a positive probability vector and the same cover has mass  $x \leq \bar{x}$ . Thus  $p$  is an interior maximizer on this affine section. The linear term vanishes, so  $z^\top M p = 0$ , and

$$e(H, p + tz) = e(H, p) + \frac{t^2}{2} z^\top M z.$$

A positive quadratic term contradicts maximality. If the quadratic term is zero, move along the line until a mass first becomes zero. Edge density and cover mass remain unchanged, producing a counterexample with smaller positive support. This is also impossible.  $\square$

**Corollary 4.4.** *The matrix  $M$  has at most two nonnegative eigenvalues. In particular,  $\lambda_3(M) < 0$ .*

*Proof.* The subspace orthogonal to  $\mathbf{1}$  and  $c$  has codimension at most two and the quadratic form is negative definite there. The min-max principle gives the assertion.  $\square$

We also record first-order stationarity. Although  $c$  was selected after the maximizing triple,  $p$  remains a maximizer with  $(H, c)$  fixed: every nearby  $p'$  satisfying  $\langle p', \mathbf{1} \rangle = 1$  and  $\langle p', c \rangle \leq \bar{x}$  makes  $(H, p', c)$  an admissible triple. Because  $p$  lies in the relative interior of the probability simplex, the normal cone is generated only by the probability equality and the possibly active cover halfspace. Hence there are  $\lambda \in \mathbb{R}$  and  $\mu \geq 0$  (with  $\mu = 0$  if that halfspace is inactive) such that

$$Mp = \lambda \mathbf{1} + \mu c. \quad (17)$$

**Lemma 4.5.** *The independence number of  $H$  is at most two.*

*Proof.* If three atoms were pairwise nonadjacent, their coordinate subspace would contain a nonzero vector orthogonal to  $\mathbf{1}$  and  $c$ , and its adjacency quadratic form would be zero, contradicting Lemma 4.3.  $\square$

**Lemma 4.6.** *The complement  $J = \overline{H}$  is a chain graph. Equivalently,  $H$  is a co-chain graph.*

*Proof.* By Lemma 4.5,  $J$  is triangle-free. If  $J$  were not bipartite, a shortest odd cycle would be an induced  $C_m$  with odd  $m \geq 5$ . Hence  $H$  would contain the induced complement  $\overline{C}_m$ . The adjacency matrix of  $\overline{C}_m$  has principal eigenvalue  $m - 3$  and two further eigenvalues equal to

$$-1 + 2 \cos(\pi/m) > 0.$$

Interlacing would give  $\lambda_3(M) > 0$ , contrary to Corollary 4.4.

Nor can  $J$  contain an induced  $2K_2$ , whose complement is an induced  $C_4$  in  $H$ . Since  $\lambda_3(C_4) = 0$ , interlacing would give  $\lambda_3(M) \geq 0$ . Finally, a bipartite graph with no induced  $2K_2$  is a chain graph: incomparable neighborhoods on one side immediately create an induced  $2K_2$ .  $\square$

**Lemma 4.7** ( $P_4$  sign rule). *If  $a - b - c - d$  is an induced  $P_4$  in  $H$ , then*

$$(c_c - c_a)(c_d - c_b) > 0.$$

*Proof.* Put  $\alpha = c_c - c_a$  and  $\beta = c_d - c_b$ . If  $(\alpha, \beta) \neq (0, 0)$ , the vector supported on the path with coordinates  $(\beta, -\alpha, -\beta, \alpha)$  is orthogonal to  $\mathbf{1}$  and  $c$ , and its quadratic value is  $-2\alpha\beta$ . Strict negativity gives  $\alpha\beta > 0$ . If  $\alpha = \beta = 0$ , the vector  $(1, 1, -1, -1)$  is orthogonal to  $\mathbf{1}$  and  $c$  and has quadratic value  $2 > 0$ , again a contradiction.  $\square$

**Lemma 4.8** (Canonical co-chain representation). *There is a possibly empty universal clique  $U$  and, unless  $H$  is complete, nonempty cliques*

$$L_0, L_1, \dots, L_{m-1} \quad \text{and} \quad R_1, R_2, \dots, R_m$$

*such that, with  $L = \bigsqcup_i L_i$  and  $R = \bigsqcup_j R_j$ ,*

$$L_i R_j \subseteq E(H) \iff i \geq j. \tag{18}$$

*The graph  $H$  is complete exactly when  $m = 0$ .*

*Proof.* Remove the isolated vertices of the chain graph  $\overline{H}$  and put them in  $U$ ; these are exactly the universal vertices of  $H$ . If vertices remain, fix a bipartition  $L \sqcup R$ . Merge equal-neighborhood vertices in  $L$  and order the resulting neighborhoods strictly by reverse inclusion. Taking successive differences on the  $R$  side yields the displayed nonempty classes, with  $N_{\overline{H}}(L_i) = R_{i+1} \cup \dots \cup R_m$ , which is equivalent to (18).  $\square$

**Lemma 4.9** (Far-corner lemma). *In the selected counterexample the Ferrers staircase has at most one step:  $m \leq 1$ .*

*Proof.* Assume  $m \geq 2$ . For  $1 \leq i \leq m$ , define

$$\mathcal{D}_i = \{c_r - c_\ell : r \in R_i, \ell \in L_{i-1}\}.$$

For  $1 \leq i < m$ , arbitrary  $\ell_{i-1} \in L_{i-1}$ ,  $\ell_i \in L_i$ ,  $r_i \in R_i$ , and  $r_{i+1} \in R_{i+1}$  induce the path

$$\ell_{i-1} - \ell_i - r_i - r_{i+1}.$$

By Lemma 4.7, every element of  $\mathcal{D}_i$  has positive product with every element of  $\mathcal{D}_{i+1}$ . Hence none of these sets contains zero and they all have the same strict sign.

Choose  $u \in L_0$  and  $v \in R_m$ . Their common neighborhood is exactly  $U$ . By Lemma 4.2, there is  $Q \cong K_{r-2}$  in  $U$  such that

$$c_u + c_v + c(Q) < 1. \tag{19}$$

If all  $\mathcal{D}_i$  are positive, choose  $w \in L_{m-1}$ . Since  $Q \cup \{u, w\}$  is a  $K_r$ , its cover inequality and (19) give  $c_w > c_v$ , whereas positivity of  $\mathcal{D}_m$  gives  $c_v > c_w$ . If all  $\mathcal{D}_i$  are negative, choose  $z \in R_1$  and use the clique  $Q \cup \{z, v\}$  to obtain  $c_z > c_u$ , while negativity of  $\mathcal{D}_1$  gives  $c_z < c_u$ .  $\square$

**Corollary 4.10.** *The support  $H$  is complete or has a partition*

$$V(H) = A \sqcup B \sqcup C \quad (20)$$

*into nonempty cliques such that  $B$  is complete to  $A \cup C$  and  $A$  is anticomplete to  $C$ . Moreover,  $|B| \geq r - 2$ .*

*Proof.* The case  $m = 0$  is complete. When  $m = 1$ , take  $A = L_0$ ,  $C = R_1$ , and  $B = U$ . For a nonedge  $uv$  with  $u \in A$ ,  $v \in C$ , the saturation certificate supplies a  $K_{r-2}$  in their common neighborhood  $B$ .  $\square$

## 5. COMPLETE SUPPORTS AND TERMINAL ZERO SETS

**5.1. Complete supports.** Suppose first that  $H$  is complete. Order the masses as  $p_1 \geq p_2 \geq \dots$  and set  $p_j = 0$  beyond the support.

**Lemma 5.1** (Finite hypersimplex formula). *For a complete support,*

$$\text{til}_r(H, p) = \min_{0 \leq s \leq r-1} \frac{1 - \sum_{j=1}^s p_j}{r - s}.$$

*Proof.* If a fractional tiling has size  $q$  and loads  $y_j$ , then each clique uses an atom at most once, whence

$$0 \leq y_j \leq \min\{p_j, q\}, \quad \sum_j y_j = rq.$$

Thus  $\sum_j \min\{p_j, q\} \geq rq$ . Conversely, if this inequality holds and  $q > 0$ , choose  $0 \leq y_j \leq \min\{p_j, q\}$  with total  $rq$ . The vector  $(y_j/q)_j$  belongs to the hypersimplex

$$\{z : 0 \leq z_j \leq 1, \sum_j z_j = r\},$$

which is the convex hull of incidence vectors of  $r$ -subsets. A convex decomposition supplies a fractional  $K_r$ -tiling of size  $q$ . For the ordered sequence  $(p_j)$ , the displayed inequality is equivalent to

$$q \leq \frac{1 - \sum_{j=1}^s p_j}{r - s} \quad (0 \leq s \leq r - 1),$$

which proves the formula.  $\square$

**Proposition 5.2.** *If  $H$  is complete and  $x = \text{fcov}_r(H, p)$ , then  $e(H, p) \leq F_r(x)$ .*

*Proof.* If  $x = 1/r$ , the result follows from  $e(H, p) \leq 1/2 = F_r(1/r)$ . Otherwise (10) and Lemma 5.1 give some  $s \in \{1, \dots, r - 1\}$  such that

$$1 - \sum_{j=1}^s p_j \leq (r - s)x.$$

Put  $i = r - s$ . Then  $\sum_{j=1}^{r-i} p_j \geq 1 - ix$ , and Cauchy-Schwarz yields

$$\sum_j p_j^2 \geq \frac{(1 - ix)^2}{r - i}.$$

Therefore

$$e(H, p) = \frac{1}{2} \left( 1 - \sum_j p_j^2 \right) \leq B_{r,i}(x) \leq F_r(x).$$

$\square$

The selected counterexample must therefore have the partition (20). Every part is a module, and  $|B| \geq r - 2$ .

### 5.2. A weighted extension of Turán's theorem.

**Theorem 5.3** (Weighted extension of Turán). *Let  $r \geq 3$ , let  $(G, p)$  be a weighted graph, and let  $Z \subseteq V(G)$  have mass  $z$ . If no  $K_r$  intersects  $Z$ , then*

$$e(G, p) \leq \begin{cases} \frac{r-2}{2(r-1)}, & z \geq 1/(r-1), \\ \frac{1}{2} - z + \frac{r-1}{2}z^2, & 0 \leq z \leq 1/(r-1). \end{cases}$$

*Proof.* For rational masses, blow up each atom to an independent class with the corresponding proportional size. The marked-set hypothesis is preserved. Applying [1, Theorem 2(a)], dividing by the square of the blow-up order, and passing to the limit gives the displayed inequality. For general  $p$ , choose positive rational probability vectors  $p^{(N)} \rightarrow p$ . The support graph and the marked-set hypothesis do not change, while  $e(G, p^{(N)}) \rightarrow e(G, p)$  and  $p^{(N)}(Z) \rightarrow z$ . The two displayed branches agree at  $z = 1/(r-1)$ , so their common piecewise function is continuous; passing to the limit proves the general case.  $\square$

Let  $Z = \{v : c_v = 0\}$ .

**Lemma 5.4** (Zero-set mass). *One has  $p(Z) \geq 1 - rx$ .*

*Proof.* Choose an optimal fractional tiling  $t$  dual to  $c$  and let  $L(v) = \sum_{K \ni v} t_K$ . Complementary slackness gives  $c_v > 0 \Rightarrow L(v) = p_v$ . Hence

$$p(V(H) \setminus Z) = \sum_{c_v > 0} L(v) \leq \sum_v L(v) = r \sum_K t_K = rx.$$

$\square$

Call  $Z$  *terminal* if no  $K_r$  intersects it.

**Lemma 5.5** (Terminal case). *If  $Z$  is terminal, then  $e(H, p) \leq F_r(x)$ .*

*Proof.* Put  $z = p(Z)$  and  $\ell = 1 - rx$ . By Lemma 5.4,  $z \geq \ell$ . If  $z \geq 1/(r-1)$ , then Theorem 5.3 gives

$$e(H, p) \leq \frac{r-2}{2(r-1)} \leq B_{r,1}(x).$$

If  $z < 1/(r-1)$ , then  $\ell < 1/(r-1)$  and  $R_r$  is defined. The function  $g(z) = 1/2 - z + (r-1)z^2/2$  is decreasing on this interval, so

$$e(H, p) \leq g(z) \leq g(\ell) = R_r(x).$$

$\square$

Every atom of  $B$  is universal. Therefore Lemma 3.4 implies

$$c_b > 0 \quad (b \in B) \tag{21}$$

in the selected counterexample.

## 6. EXTREME COVERS ON A ONE-STEP SUPPORT

For  $S \subseteq V(H)$  with  $|S| \geq r$ , put

$$m_S(c) = \min\{c(K) : K \subseteq S, |K| = r\}.$$

Call  $S$  *active* if  $m_S(c) = 1$ . The two complete sides are  $S_A = A \cup B$  and  $S_C = B \cup C$ .

**Lemma 6.1** (Tight sets use the lowest levels). *Let  $S \in \{S_A, S_C\}$  and let  $K \subseteq S$  be a tight  $r$ -set. If  $u \in S \setminus K$  and  $v \in K$ , then  $c_u \geq c_v$ . Consequently all tight  $r$ -sets in  $S$  use the same number of vertices at every cover level.*

*Proof.* If  $c_u < c_v$ , replacing  $v$  by  $u$  gives an  $r$ -set in the same complete side with cover sum below one. Thus a tight set consists of the  $r$  lowest available cover values, with freedom only inside the threshold level.  $\square$

**Lemma 6.2** (Two-level lemma). *The extreme minimum cover  $c$  has at most two distinct positive coordinate values. If it has exactly two, the active sides contribute linearly independent level-count equations.*

*Proof.* Let  $0 < \theta_1 < \dots < \theta_t$  be the positive values of  $c$ . For each active side  $S$ , let  $a_S \in \mathbb{R}^t$  count the numbers of the positive levels in a tight  $r$ -set; this is well defined by Lemma 6.1. Every  $K_r$  is contained in one of the two complete sides, so every tight constraint is represented by one of these active count vectors. If  $t > 2$ , choose a nonzero vector orthogonal to the at most two active count vectors and perturb every coordinate at level  $\theta_j$  by plus or minus a sufficiently small common multiple of its  $j$ th entry. Positivity and the strict order of levels are preserved, tight constraints remain equalities, and the finitely many non-tight constraints remain feasible. The two perturbed covers have  $c$  as their midpoint, contradicting extremality. The same perturbation when  $t = 2$  shows that the active count vectors must span  $\mathbb{R}^2$ .  $\square$

Order the  $B$ -coordinates as  $b_1 \leq \dots \leq b_{m_B}$ , where  $m_B = |B| \geq r - 2$ , and put

$$\sigma = \sum_{i=1}^{r-2} b_i. \quad (22)$$

For every  $a \in A$  and  $z \in C$ , the saturation certificate for the nonedge  $az$  gives an  $(r - 2)$ -clique  $Q \subseteq B$  with  $c_a + c_z + c(Q) < 1$ . Since  $\sigma \leq c(Q)$ , it follows that

$$c_a + c_z + \sigma < 1 \quad (a \in A, z \in C). \quad (23)$$

**Lemma 6.3.** *Each of  $A$  and  $C$  contains at most one zero-cover atom.*

*Proof.* If  $a_1, a_2 \in A$  were distinct zero-cover atoms, then  $\{a_1, a_2\}$  together with the  $r - 2$  lightest atoms of  $B$  would be a  $K_r$ , giving  $\sigma \geq 1$ , contrary to (23). The proof for  $C$  is symmetric.  $\square$

**Theorem 6.4** (Rigid cover classification). *For the selected counterexample with partition (20) and extreme minimum cover  $c$ , assume (21). After possibly interchanging  $A$  and  $C$ , one of the following holds.*

- (i) *No  $K_r$  intersects the zero set  $Z$ .*
- (ii) *The sets  $A$  and  $C$  each consist of one zero-cover atom, and*

$$c_b = \frac{1}{r-1} \quad (b \in B).$$

- (iii) *The set  $C$  consists of one zero-cover atom. There are integers*

$$a_0 = |A|, \quad 0 \leq s \leq r - 2, \quad t \geq 0, \quad j = r - 1 - s \geq 1$$

*such that  $c \equiv 1/r$  on  $A$ , exactly  $s$  atoms of  $B$  have cover value  $1/r$ , and its remaining  $t$  atoms have value*

$$\gamma = \frac{j+1}{rj}.$$

*Moreover,  $a_0 \geq j + 1$  and  $t \geq j$ .*

*Proof.* We divide the proof into four steps.

*Step 1:*  $m_B = r - 2$ . If  $|A|, |C| \geq 2$ , let  $a_1, a_2$  and  $z_1, z_2$  be the two lightest atoms of the respective ends. The cliques consisting of either pair together with  $B$  give

$$c_{a_1} + c_{a_2} + \sigma \geq 1, \quad c_{z_1} + c_{z_2} + \sigma \geq 1.$$



Thus  $c_{a_2}, c_{z_2} \geq (1 - \sigma)/2$ , contradicting (23) for  $a_2, z_2$ . Hence one end is a singleton; say  $C = \{z\}$ . The side  $B \cup C$  has only  $r - 1$  atoms, so  $z$  lies in no  $K_r$  and optimality gives  $c_z = 0$ . If  $A$  contained a zero atom  $a$  and another atom  $a'$ , then  $\{a, a'\} \cup B$  would give  $c_{a'} + \sigma \geq 1$ , again contradicting (23). Hence no  $K_r$  meets  $Z$ , which is case (i).

*Step 2: preliminary consequences for  $m_B \geq r - 1$ .* Put  $\beta = b_{r-1}$ . The  $r - 1$  lightest atoms of  $B$ , together with any  $a \in A$  or  $z \in C$ , form a  $K_r$ . Thus

$$c_a + \sigma + \beta \geq 1, \quad c_z + \sigma + \beta \geq 1.$$

Together with (23), this gives

$$c_a < \beta, \quad c_z < \beta. \quad (24)$$

Suppose that an end atom has positive cover value. Then Lemma 6.2 and (24) show that the positive levels are  $0 < \alpha < \gamma$ , every positive end atom has value  $\alpha$ , and  $\beta = \gamma$ . Let

$$s = |\{b \in B : c_b = \alpha\}| \leq r - 2.$$

For an end  $E \in \{A, C\}$ , let  $z_E \in \{0, 1\}$  be its number of zero atoms and  $n_E$  its number of positive atoms. If  $E \cup B$  is active, its tight equation is

$$(r - z_E)\alpha = 1, \quad \text{if } n_E + s \geq r - z_E, \quad (25)$$

$$(n_E + s)\alpha + (r - z_E - n_E - s)\gamma = 1, \quad \text{if } n_E + s < r - z_E. \quad (26)$$

*Step 3: both ends cannot contain positive atoms.* Assume  $n_A, n_C \geq 1$ . Both sides are active by Lemma 4.1. If both are in the  $\alpha$ -threshold case, no tight clique contains a  $\gamma$ -atom of  $B$ , contradicting Lemma 4.1. Suppose that  $A \cup B$  is in the  $\alpha$ -threshold case and  $B \cup C$  in the  $\gamma$ -threshold case. Comparing their equations gives

$$(z_C - z_A)\alpha = (r - z_C - n_C - s)(\gamma - \alpha) > 0.$$

Thus  $z_A = 0$ ,  $z_C = 1$ , and  $\alpha = 1/r$ . Put  $q = n_C \geq 1$  and  $j = r - 1 - q - s \geq 1$ . Then  $\gamma = (j + 1)/(rj)$ , and

$$2\alpha + \sigma = \frac{s + 2}{r} + (r - 2 - s)\frac{j + 1}{rj} = 1 + \frac{j + q - 1}{rj} > 1,$$

contrary to (23). The reverse assignment of thresholds is symmetric.

If both sides are in the  $\gamma$ -threshold case and  $z_A = z_C$ , their equations give  $n_A = n_C$ , so the active level-count equations coincide, contrary to Lemma 6.2. If, say,  $z_A = 0$ ,  $z_C = 1$ , subtraction gives

$$(n_A - n_C)(\gamma - \alpha) = \gamma.$$

In particular  $n_A \geq 2$ , and comparison of the  $A \cup B$  tight equation with  $2\alpha + \sigma$  yields

$$2\alpha + \sigma - 1 = (n_A - 2)(\gamma - \alpha) \geq 0,$$

again contradicting (23).

*Step 4: one end is entirely zero.* After interchanging the ends, assume  $C = \{z\}$  and  $c_z = 0$ . If  $A$  contained both a zero and a positive atom, then it would have one zero atom and all positive atoms would have value  $\alpha$ . The side  $A \cup B$  is active. In its  $\alpha$ -threshold case,  $(r - 1)\alpha = 1$ . A  $\gamma$ -atom of  $B$  must then be tight in  $B \cup C$ , whose equation is

$$s\alpha + (r - 1 - s)\gamma = 1,$$

forcing  $\gamma = \alpha$ . In the  $\gamma$ -threshold case, if  $B \cup C$  is active, subtraction of the two equations gives  $n_A(\alpha - \gamma) = 0$ ; if it is inactive, only one active count equation remains for two positive levels. Both alternatives are impossible.

If every atom of  $A$  is zero, then  $A = \{a\}$  by Lemma 6.3. The two sides have the same multiset of cover values. At least one is active by Lemma 4.1, hence both are active. Two positive levels

would give identical active count equations, so every atom of  $B$  has a common value  $\lambda$ . A tight clique uses one zero end and  $r - 1$  atoms of  $B$ , whence  $(r - 1)\lambda = 1$ . This is case (ii).

Finally suppose that all  $a_0 = |A|$  atoms of  $A$  are positive. If  $a_0 + s \geq r$ , then  $A \cup B$  is in the  $\alpha$ -threshold case, so  $\alpha = 1/r$ . The  $\gamma$ -atoms of  $B$  force  $B \cup C$  to be active, and

$$s\alpha + (r - 1 - s)\gamma = 1.$$

With  $j = r - 1 - s$ , this gives  $\gamma = (j + 1)/(rj)$ ,  $a_0 \geq j + 1$ , and  $t = m_B - s \geq j$ , yielding case (iii).

It remains to rule out  $a_0 + s < r$ . Then  $A \cup B$  is in the  $\gamma$ -threshold case. The other side must be active, since otherwise there is only one active count equation, and subtraction gives

$$a_0\alpha + (1 - a_0)\gamma = 0.$$

For  $a_0 = 1$  this says  $\alpha = 0$ . For  $a_0 \geq 2$ , make both signed perturbations obtained by adding  $\varepsilon$  to one of two  $A$ -coordinates and subtracting it from the other. Every tight set in  $A \cup B$  contains all of  $A$ , while tight sets in  $B \cup C$  contain none of  $A$ ; all tight sums are unchanged. A sufficiently small perturbation also preserves every non-tight constraint. The two resulting covers are distinct and have  $c$  as their midpoint, contradicting the extremality of  $c$ .  $\square$

**6.1. The two rigid nonterminal types.** Write

$$\Delta(H, p) = 1 - 2e(H, p). \quad (27)$$

For a simple weighted support this is the sum of the squared atom masses plus twice the mass of the nonedges between distinct atoms.

**Proposition 6.5.** *In case (ii) of Theorem 6.4,  $e(H, p) \leq B_{r,r-1}(x)$ .*

*Proof.* Put  $b = p(B)$ . Then  $x = b/(r - 1)$  and  $p(A) + p(C) = 1 - (r - 1)x$ . Since the only cross-module nonedges lie between  $A$  and  $C$ ,

$$\Delta(H, p) = p(A)^2 + p(C)^2 + 2p(A)p(C) + \sum_{v \in B} p_v^2 \geq (1 - (r - 1)x)^2.$$

This is equivalent to the desired bound.  $\square$

For the rest of this subsection,  $(H, p, c)$  remains the selected extremal triple, and assume case (iii). Thus  $C = \{z\}$  with  $c_z = 0$ , all  $a_0 = |A|$  atoms of  $A$  have cover value  $1/r$ , and  $B$  has  $s$  low atoms of value  $1/r$  and  $t$  high atoms of value  $\gamma = (j + 1)/(rj)$ , where  $j = r - 1 - s$ ,  $a_0 \geq j + 1$ , and  $t \geq j$ . Put

$$N = a_0 - 1, \quad h = N - j \geq 0, \quad \kappa = t - j \geq 0. \quad (28)$$

Atoms in the same module and at the same cover level have equal masses: subtracting their equations in (17) makes the degree difference, which is the negative of the mass difference, vanish. Write  $w$  for each  $A$ -mass,  $L$  for each low  $B$ -mass when  $s > 0$ ,  $u$  for each high  $B$ -mass, and  $d = p_z$ .

**Lemma 6.6** (KKT parametrization). *Define  $v = Nw/j$ . When  $s = 0$ , set  $L = u + v$  formally; all terms involving  $L$  are then multiplied by  $s$ . One has*

$$w = \frac{j}{N}v, \quad d = u + \frac{h}{N}v, \quad L = u + v, \quad (29)$$

and

$$rv + (r + \kappa)u = 1. \quad (30)$$

If  $\ell = 1 - rx$ , then

$$\ell = \frac{h}{N}v - \frac{\kappa}{j}u. \quad (31)$$

Finally,

$$\Delta(H, p) = (r + \kappa)u^2 + 2ruv + \left(r + \frac{jh}{N}\right)v^2. \quad (32)$$

*Proof.* The weighted degrees of an  $A$ -atom, a  $B$ -atom of mass  $q$ , and  $z$  are respectively

$$1 - d - w, \quad 1 - q, \quad p(B).$$

The total-mass equation is

$$a_0 w + sL + tu + d = 1. \quad (33)$$

Comparing an  $A$ -atom with  $z$  in (17) gives  $\mu/r = Nw$ . Comparing a high  $B$ -atom with  $z$  then gives

$$a_0 w + d - u = \mu\gamma = Nw \frac{j+1}{j},$$

so  $d - u = hw/j$ . Comparing a low  $B$ -atom with  $z$  gives  $L = d + w$ . Substitution of  $v = Nw/j$  yields (29). Substituting these expressions,  $s = r - 1 - j$ , and  $t = j + \kappa$  in (33) gives (30).

The cover mass is

$$x = \frac{a_0 w + sL}{r} + t \frac{j+1}{rj} u.$$

Subtracting  $rx$  from (30) gives (31). Finally, the missing pairs are the atom diagonals and all pairs between  $A$  and  $z$ , so

$$\Delta(H, p) = a_0 w^2 + sL^2 + tu^2 + d^2 + 2a_0 wd.$$

Substitution gives (32). □

**Proposition 6.7.** *For the selected extremal triple in the rigid two-level case (iii),  $e(H, p) \leq R_r(x)$ .*

*Proof.* Since  $\ell = 1 - rx$ , one has  $\ell \geq 0$ . The high atoms occur and have  $u > 0$ , so (30) gives  $v < 1/r$ . Therefore

$$\ell \leq \frac{h}{N}v < \frac{1}{r} < \frac{1}{r-1},$$

and  $R_r(x)$  is defined. It suffices to prove

$$\Delta(H, p) \geq 2\ell - (r-1)\ell^2. \quad (34)$$

Using  $1 = rv + (r + \kappa)u$ , the difference between the two sides is the homogeneous quadratic form

$$Au^2 + 2Buv + Cv^2,$$

where

$$\begin{aligned} A &= \frac{j^2(r + \kappa) + 2j\kappa(r + \kappa) + \kappa^2(r - 1)}{j^2}, \\ B &= \frac{jr(j + \kappa) - h\kappa(j - 1)}{j(j + h)}, \\ C &= \frac{j^2r + hj^2 + h^2(j - 1)}{(j + h)^2}. \end{aligned}$$

Clearly  $A, C > 0$ , and direct factorization gives

$$AC - B^2 = \frac{hj\kappa + hjr + h\kappa r + j\kappa r}{j^2(j + h)^2} \left( h(j - 1)(j + \kappa) + j(j^2 + \kappa(2j - 1)) \right) \geq 0.$$

Thus the quadratic form is positive semidefinite, proving (34). Since  $1 - 2R_r(x) = 2\ell - (r - 1)\ell^2$ , the result follows. □

## 6.2. Completion of the weighted induction.

*Proof of Theorem 3.1.* The case  $r = 2$  is Proposition 3.2. Let  $r \geq 3$  and assume the theorem for  $r - 1$ . If a counterexample existed, choose the extremal triple above. By Corollary 4.10, its support would be complete or a clique blow-up of  $P_3$ . The complete case is excluded by Proposition 5.2. In the latter case, if no  $K_r$  intersects the zero set, Lemma 5.5 applies; if a vertex of  $B$  has zero cover value, Lemma 3.4 applies. Otherwise Theorem 6.4 applies. Its case (i) is terminal, case (ii) is handled by Proposition 6.5, and case (iii) by Proposition 6.7. Every alternative contradicts (16).

Thus the weighted inequality holds for every  $r \geq 2$ . If  $c$  is any fractional cover, then  $\text{fcov}_r(H, p) \leq \langle p, c \rangle$ , and the final assertion follows from the monotonicity of  $F_r$ .  $\square$

## 7. GRAPHON, TILING, COVER, AND STRUCTURAL PRELIMINARIES

A graphon is a symmetric measurable function  $W : \Omega^2 \rightarrow [0, 1]$  on a standard atomless probability space. All statements are invariant under measure-space isomorphism, so whenever a localization argument is used we identify the space with  $([0, 1], \lambda)$ . We write  $\overline{W} = 1 - W$  and

$$e(W) = \frac{1}{2} \int_{\Omega^2} W, \quad d_W(x) = \int_{\Omega} W(x, y) \, dy,$$

and, for  $h \in L^\infty(\Omega)$ ,

$$Q_W(h) = \int_{\Omega^2} W(x, y) h(x) h(y) \, dx \, dy.$$

A fractional  $K_r$ -cover is a measurable  $c : \Omega \rightarrow [0, 1]$  such that  $\sum_{j=1}^r c(x_j) \geq 1$  for almost every  $K_r$  in the support of  $W$ . Restricting to  $[0, 1]$  loses nothing, since coordinates above one may be truncated. Its minimum mass is  $\text{fcov}(K_r, W)$ . A fractional  $K_r$ -tiling is a measurable function  $t : \Omega^r \rightarrow [0, \infty)$ , supported on the ordered  $K_r$ -tuples, whose load

$$\ell_t(v) = \sum_{i=1}^r \int_{\Omega^{r-1}} t(x_1, \dots, x_{i-1}, v, x_{i+1}, \dots, x_r) \, d\mu^{r-1}$$

is at most one almost everywhere. Its size is  $|t| = \int_{\Omega^r} t \, d\mu^r$ , and  $\text{til}(K_r, W)$  is the supremum of these sizes. In particular,  $\int \ell_t = r|t|$ . Throughout the paper, assertions that a measurable set meets or avoids every clique are understood modulo the corresponding product-measure null set on the clique support. Whenever a coordinate of a clique tuple is replaced, we first restrict to the full-measure set of Fubini-typical tuples on which all required cover inequalities hold.

For a bounded kernel  $U$ , the cut norm is

$$\|U\|_{\square} = \sup_{S, T \subseteq \Omega} \left| \int_{S \times T} U \right|.$$

The cut distance  $\delta_{\square}$  and the  $L^1$  distance  $\delta_1$  are obtained by optimizing over measure-preserving relabelings. For a family  $\mathcal{U}$  of graphons,  $\delta_1(W, \mathcal{U})$  and  $\delta_{\square}(W, \mathcal{U})$  denote the corresponding infima over  $U \in \mathcal{U}$ . For two graphs on the same vertex set,  $\text{edit}(G, H) = |E(G) \Delta E(H)|$ ; distance to a family also minimizes over vertex relabelings and over members of the family. After relabeling,  $2\text{edit}(G, H)/n^2$  is the  $L^1$  distance of their graphons up to  $O(1/n)$ .

**Theorem 7.1** (Graphon tiling inputs). *The graphon space modulo weak isomorphism is compact in cut distance [3, 9]. If  $W_n \rightarrow W$  in cut distance, then*

$$e(W_n) \rightarrow e(W), \quad \text{til}(K_r, W) \leq \liminf_n \text{til}(K_r, W_n).$$

*A minimum fractional cover exists and*

$$\text{til}(K_r, W) = \text{fcov}(K_r, W).$$

More precisely, lower semicontinuity, cover compactness and attainment, tiling-cover duality, and finite-to-graphon transference are [7, Theorems 3.7, 3.14, and 3.16, equation (3.7), and Theorem 3.4]. In particular, if finite graphs  $G_n$  converge to  $W$  in cut distance, then

$$\text{til}(K_r, W) \leq \liminf_{n \rightarrow \infty} \frac{\nu(K_r, G_n)}{|V(G_n)|}. \quad (35)$$

*Proof of Theorem 1.2.* Let  $x = \text{til}(K_r, W) = \text{fcov}(K_r, W)$  and choose a minimum fractional cover  $c : \Omega \rightarrow [0, 1]$  of mass  $x$ , using Theorem 7.1. If  $x = 1/r$ , then  $e(W) \leq 1/2 = F_r(1/r)$ , so suppose  $x < 1/r$ .

Take independent samples  $X_1, X_2, \dots$  from  $\mu$ . Conditional on the points, join every pair  $i < j$  independently with probability  $W(X_i, X_j)$ , and let  $H_m$  be the graph induced on  $[m]$ . With probability one,

$$\frac{e(H_m)}{m^2} \rightarrow e(W), \quad x_m := \frac{1}{m} \sum_{i=1}^m c(X_i) \rightarrow x.$$

Moreover, almost surely the sampled vector  $(c(X_1), \dots, c(X_m))$  is a fractional  $K_r$ -cover of every  $H_m$ . Indeed, for a fixed ordered  $r$ -tuple of distinct indices, the probability that it is a realized clique and violates the cover inequality is

$$\int_{\Omega^r} \mathbf{1}_{\{\sum_{a=1}^r c(x_a) < 1\}} \prod_{a < b} W(x_a, x_b) \, d\mu^r = 0.$$

A countable union over index tuples remains null.

For all sufficiently large  $m$ ,  $x_m < 1/r$ . Apply Theorem 3.1 to the uniformly weighted graph  $H_m$  and this sampled cover:

$$\frac{e(H_m)}{m^2} \leq F_r(x_m).$$

Letting  $m \rightarrow \infty$  and using the continuity of  $F_r$  (immediate from its two-branch definition for  $r = 2$ , and supplied by Lemma 2.3 for  $r \geq 3$ ) proves the theorem.  $\square$

*Proof of Theorem 1.1.* The lower bound is Proposition 2.4. If the upper bound failed uniformly, there would be  $\varepsilon > 0$ , integers  $n_j \rightarrow \infty$  and  $0 \leq k_j \leq \lfloor n_j/r \rfloor$ , and  $n_j$ -vertex graphs  $G_j$  such that

$$\nu(K_r, G_j) \leq k_j, \quad e(G_j) \geq (F_r(k_j/n_j) + \varepsilon)n_j^2.$$

After taking a subsequence, assume  $k_j/n_j \rightarrow x$  and  $W_{G_j} \rightarrow W$  in cut distance. The finite-to-graphon transference in Theorem 7.1 gives

$$\text{til}(K_r, W) \leq \liminf_{j \rightarrow \infty} \frac{\nu(K_r, G_j)}{n_j} \leq x.$$

Consequently, by Theorem 1.2 and the monotonicity of  $F_r$ ,

$$e(W) \leq F_r(\text{til}(K_r, W)) \leq F_r(x).$$

On the other hand, cut convergence of the edge density and continuity of  $F_r$  give

$$e(W) = \lim_{j \rightarrow \infty} \frac{e(G_j)}{n_j^2} \geq F_r(x) + \varepsilon,$$

a contradiction.  $\square$

**Theorem 7.2** (Graphon extension of Turán: inequality and equality). *Let  $r \geq 3$ , and let  $Z$  be a measurable set such that the set of  $K_r$ -tuples of  $W$  having at least one coordinate in  $Z$  has*

$\mu^r$ -measure zero. Put  $z = \mu(Z)$ . Then

$$e(W) \leq g_r(z) := \begin{cases} \frac{r-2}{2(r-1)}, & z \geq 1/(r-1), \\ \frac{1}{2} - z + \frac{r-1}{2}z^2, & 0 \leq z \leq 1/(r-1). \end{cases} \quad (36)$$

If equality holds and  $z \geq 1/(r-1)$ , then  $W$  is the balanced complete  $(r-1)$ -partite graphon. If equality holds and  $z \leq 1/(r-1)$ , then  $W$  is weakly isomorphic to an extension-of-Turán graphon. More precisely,  $W$  and that graphon admit a common pullback on a probability space with a partition

$$\Omega = X \sqcup \bigcup_{j=1}^{r-1} (Y_j \sqcup Z_j)$$

with

$$\mu(X) = 1 - (r-1)z, \quad \mu(Y_j) + \mu(Z_j) = z, \quad \sum_{j=1}^{r-1} \mu(Z_j) = z.$$

Writing  $Z'$  for the corresponding pullback of the originally prescribed set  $Z$ , one has  $Z' = \bigsqcup_{j=1}^{r-1} Z_j$  modulo null sets. On the common pullback,  $X$  is a clique, each  $Y_j \sqcup Z_j$  is independent, distinct such classes are completely joined, and  $X$  is complete to  $Y_j$  and anticomplete to  $Z_j$ .

*Proof.* We derive the graphon statement from the finite equality and stability theorem of Allen–Böttcher–Hladký–Piguet [1]. Sample independent points  $X_1, X_2, \dots$  from  $\mu$  and, conditional on them, form the usual  $W$ -random graph  $G_n$  on  $[n]$ . Put

$$M_n = \{i \in [n] : X_i \in Z\}, \quad m_n = |M_n|.$$

With probability one,

$$\frac{m_n}{n} \longrightarrow z, \quad \frac{e(G_n)}{n^2} \longrightarrow e(W), \quad W_{G_n} \longrightarrow W \quad \text{in cut distance.}$$

Moreover, almost surely no copy of  $K_r$  in any  $G_n$  intersects  $M_n$ . Indeed, for each fixed ordered  $r$ -tuple of distinct indices, the probability that it realizes a clique and that at least one sampled point lies in  $Z$  is

$$\int_{\Omega^r} \mathbf{1}_{\{\{x_1, \dots, x_r\} \cap Z \neq \emptyset\}} \prod_{a < b} W(x_a, x_b) \, d\mu^r = 0,$$

and a countable union over index tuples is still null.

Let  $t_{r-1}(n, m)$  and  $\mathfrak{E}_{r-1}(n, m)$  denote, respectively, the extremal edge number and the extremal family in [1, Theorem 2]. The finite extremal theorem therefore gives

$$e(G_n) \leq t_{r-1}(n, m_n),$$

where

$$\frac{t_{r-1}(n, m)}{n^2} = g_r(m/n) + O_r(1/n)$$

uniformly in  $m$ . Letting  $n \rightarrow \infty$  proves (36).

Now assume equality. Choose a sequence  $\varepsilon_s \downarrow 0$ . The finite stability theorem supplies  $\gamma_s > 0$  for  $\varepsilon_s$ . Along an almost-sure sample path, choose  $n_s \rightarrow \infty$  so large that

$$t_{r-1}(n_s, m_{n_s}) - e(G_{n_s}) < \gamma_s n_s^2.$$

The marked-set stability statement [1, Theorem 2(c)] gives an extremal graph  $H_s \in \mathfrak{E}_{r-1}(n_s, m_{n_s})$  on the same vertex set, with the marked set  $M_{n_s}$  unchanged, and

$$\text{edit}(G_{n_s}, H_s) \leq \varepsilon_s n_s^2.$$

If  $z > 1/(r-1)$ , these extremal graphs have the balanced Turán graphon as their only limit, which proves the asserted equality statement in that range. Suppose now that  $z \leq 1/(r-1)$ . In the finite description, only  $O_r(1)$  vertices are sporadic and therefore disappear in the graphon limit. The remaining vertices consist of a clique of asymptotic measure  $1 - (r-1)z$  and  $r-1$  independent classes of asymptotic measure  $z$ ; the marked set  $M_{n_s}$  is distributed among those classes. Relabel the  $r-1$  classes consistently and pass to a subsequence on which all normalized class and marked-class intersection sizes converge. Their marked limits are numbers  $z_1, \dots, z_{r-1} \in [0, z]$  with  $\sum_j z_j = z$ . The resulting step graphons then converge in  $L^1$  to an extension-of-Turán graphon  $U$  with a partition

$$\Omega = X \sqcup \bigcup_{j=1}^{r-1} (Y_j \sqcup Z_j)$$

and with all the adjacency and measure relations displayed in the theorem. At the threshold  $z = 1/(r-1)$  the two finite extremal regimes have the same graphon limit, namely the balanced complete  $(r-1)$ -partite graphon. Since

$$\delta_1(W_{G_{n_s}}, W_{H_s}) \leq 2\varepsilon_s + o(1)$$

and  $W_{G_{n_s}} \rightarrow W$  in cut distance,  $W$  is weakly isomorphic to  $U$ .

It remains to identify the originally prescribed set  $Z$ . By the standard common-pullback characterization of weak isomorphism, pull  $W$  and  $U$  back to one common probability space, and denote the corresponding pullback of  $Z$  by  $Z'$ , reusing  $X, Y_j, Z_j$  for the pullbacks of the displayed blocks. Then  $\mu(Z') = z$  and the set of  $K_r$ -tuples of  $U$  meeting  $Z'$  has measure zero. If  $z < 1/(r-1)$ , then  $\mu(X) > 0$ . Every positive-measure subset of  $X$  lies in a positive-measure family of cliques contained in  $X^r$ , and every positive-measure subset of a  $Y_j$  lies in a positive-measure family of cliques obtained by adjoining  $r-1$  points of  $X$ . Hence

$$Z' \subseteq \bigcup_{j=1}^{r-1} Z_j \quad \text{modulo null sets.}$$

Both sides have measure  $z$ , so equality holds modulo null sets. If  $z = 1/(r-1)$ , then  $X$  is null and  $U$  is the balanced Turán graphon with parts  $P_1, \dots, P_{r-1}$ ; define

$$Z_j = Z' \cap P_j, \quad Y_j = P_j \setminus Z_j.$$

This gives the required marked decomposition also at the threshold.  $\square$

The next lemma replaces every appeal to an attained optimal tiling.

**Lemma 7.3** (Asymptotic complementary slackness). *Let  $c$  be a minimum fractional  $K_r$ -cover of  $W$ , of mass  $x$ . There are fractional tilings  $t_n$  of sizes  $|t_n| \rightarrow x$  with load functions  $\ell_n : \Omega \rightarrow [0, 1]$  such that, writing*

$$\sigma_c(x_1, \dots, x_r) = \sum_{j=1}^r c(x_j) - 1$$

*on the clique support, one has*

$$x - |t_n| = \int_{\Omega} c(1 - \ell_n) + \int_{\Omega^r} t_n \sigma_c \longrightarrow 0. \quad (37)$$

*Consequently:*

- (i) if  $P = \{c > 0\}$ , then  $\mu(P) \leq rx$ ;
- (ii) for every  $\delta > 0$ , the  $t_n$ -mass of cliques with cover sum at least  $1 + \delta$  tends to zero;
- (iii) if every clique of cover sum below  $1 + \delta$  meets a measurable set  $S$ , then  $\mu(S) \geq x$ ;
- (iv) if every such clique avoids  $S$ , then  $\int_S \ell_n \rightarrow 0$ ;
- (v) if, in addition,  $c \geq \gamma > 0$  on  $S$ , then in the situation of (iv) one has  $\mu(S) = 0$ .

*Proof.* Choose tilings with  $|t_n| \rightarrow \text{til}(K_r, W) = x$ . By the definition of the load,

$$\int_{\Omega} c \ell_n = \int_{\Omega^r} t_n \sum_{j=1}^r c(x_j).$$

Subtracting  $|t_n|$  and using  $\ell_n \leq 1$  gives the exact decomposition (37); both terms are nonnegative, so both tend to zero.

For  $P_\gamma = \{c \geq \gamma\}$ , the first term in (37) gives

$$\int_{P_\gamma} (1 - \ell_n) \rightarrow 0, \quad \int_{P_\gamma} \ell_n \rightarrow \mu(P_\gamma).$$

Since  $\int \ell_n = r|t_n| \rightarrow rx$ , we obtain  $\mu(P_\gamma) \leq rx$  and then let  $\gamma \downarrow 0$ . Assertion (ii) follows from Markov's inequality applied to the second term of (37).

For a measurable set  $S$ , let  $N_S(Q)$  be the number of coordinates of the clique tuple  $Q$  that lie in  $S$ . Fubini's theorem and the definition of the load give

$$\int_S \ell_n = \int_{\Omega^r} t_n(Q) N_S(Q) d\mu^r(Q). \quad (38)$$

Let  $\mathcal{L}_\delta$  be the set of clique tuples of cover sum below  $1 + \delta$ . By (ii), the  $t_n$ -mass of its complement tends to zero. If every tuple in  $\mathcal{L}_\delta$  meets  $S$ , modulo a null set, then  $N_S \geq 1$  there and (38) gives

$$\liminf_n \int_S \ell_n \geq \lim_n |t_n| = x.$$

Since  $\ell_n \leq 1$ , this proves (iii). If every tuple in  $\mathcal{L}_\delta$  avoids  $S$ , then

$$\int_S \ell_n \leq r \int_{\mathcal{L}_\delta^c} t_n \rightarrow 0,$$

which proves (iv). Finally, if  $c \geq \gamma$  on  $S$ , the first term of (37) gives  $\int_S (1 - \ell_n) \rightarrow 0$ ; combining this with (iv) yields  $\mu(S) = 0$  and proves (v).  $\square$

**Lemma 7.4** (Saturation of an extremizer). *If  $W \in \mathcal{M}_r(x)$ , then*

$$\text{til}(K_r, W) = \text{fcov}(K_r, W) = x,$$

*and  $W$  is 0/1-valued almost everywhere.*

*Proof.* Put  $y = \text{til}(K_r, W) \leq x$ . By (1) and the strict monotonicity in Lemma 2.3,

$$F_r(x) = e(W) \leq F_r(y) \leq F_r(x),$$

so  $y = x$ . Let  $c$  be a minimum cover and replace  $W$  by  $W^+ = \mathbf{1}_{\{W > 0\}}$ . The same  $c$  is a cover of  $W^+$ , while  $e(W^+) \geq e(W)$ . The graphon density theorem gives  $e(W^+) \leq F_r(x) = e(W)$ , hence  $W = W^+$  almost everywhere.  $\square$

**Lemma 7.5** (Density of bounded constrained vectors). *Let  $\phi_1, \dots, \phi_m \in L^\infty(\Omega)$  and put*

$$H = \{h \in L^2(\Omega) : \langle h, \phi_j \rangle = 0 \text{ for every } j\}.$$

*Then  $H \cap L^\infty(\Omega)$  is dense in  $H$ . Consequently, every continuous quadratic-form inequality proved for bounded vectors in  $H$  extends to all of  $H$ .*

*Proof.* Let  $V = \text{span}\{\phi_1, \dots, \phi_m\}$  and let  $P_V$  be the orthogonal projection of  $L^2$  onto  $V$ . If  $h \in H$ , let  $u_N$  be the truncation of  $h$  at height  $N$  and set

$$h_N = u_N - P_V u_N.$$

The space  $V$  is finite-dimensional and contained in  $L^\infty$ , so  $h_N$  is bounded and belongs to  $H$ . Since  $u_N \rightarrow h$  in  $L^2$  and  $P_V h = 0$ , the continuity of  $P_V$  gives  $h_N \rightarrow h$  in  $L^2$ .  $\square$



**Lemma 7.6** (Null vectors of a semidefinite form). *Let  $T$  be a bounded self-adjoint operator and let  $H$  be a closed subspace. If  $\langle h, Th \rangle \leq 0$  for every  $h \in H$  and equality holds for some  $h_0 \in H$ , then  $Th_0 \in H^\perp$ .*

*Proof.* For  $g \in H$ , the quadratic polynomial  $\langle h_0 + tg, T(h_0 + tg) \rangle$  is nonpositive for both signs of  $t$  and has zero constant term. Its linear coefficient is therefore zero.  $\square$

**Lemma 7.7** (Row constancy from annihilation). *Let  $K \in L^2(\Omega^2)$  be symmetric, and let  $C \subseteq \Omega$  have positive measure. Suppose that*

$$T_K h = 0$$

*for every bounded  $h$  supported on  $C$  with  $\int_C h = 0$ . Then there is a measurable function  $g : \Omega \rightarrow \mathbb{R}$  such that*

$$K(x, y) = g(y)$$

*for almost every  $(x, y) \in C \times \Omega$ . If  $K$  is 0/1-valued on  $C \times \Omega$ , then  $g$  is 0/1-valued almost everywhere.*

*Proof.* The bounded mean-zero functions are dense in  $L_0^2(C) = \{h \in L^2(C) : \int_C h = 0\}$  by Lemma 7.5. Choose a countable dense sequence  $(h_n)$  of bounded functions in  $L_0^2(C)$ . By Fubini, after deleting one null set of  $y$ -values, the section  $K(y, \cdot)|_C$  belongs to  $L^2(C)$  and

$$\int_C K(y, x) h_n(x) dx = 0 \quad \text{for every } n.$$

Thus  $K(y, \cdot)|_C$  is orthogonal to  $L_0^2(C)$  and is constant almost everywhere on  $C$ . Taking

$$g(y) = \frac{1}{\mu(C)} \int_C K(x, y) dx$$

and using symmetry gives the displayed conclusion. The final assertion follows because an almost-everywhere constant function with values in  $\{0, 1\}$  has constant value in  $\{0, 1\}$ .  $\square$

**Lemma 7.8** (Rectangle localization). *Let  $E \subseteq [0, 1]^k$  be measurable with positive measure. For every  $\xi > 0$  there are pairwise disjoint intervals  $I_1, \dots, I_k$  of one common positive length such that*

$$\frac{\lambda^k(E \cap (I_1 \times \dots \times I_k))}{\prod_{j=1}^k \lambda(I_j)} > 1 - \xi.$$

*The same conclusion holds on any standard atomless probability space after transporting the data to  $[0, 1]$ .*

*Proof.* Almost every point of  $E$  is a Lebesgue density point. The set of tuples with a repeated coordinate has measure zero, so choose a density point  $(x_1, \dots, x_k) \in E$  with distinct coordinates. Take disjoint neighborhoods of the  $x_j$  and then equal-length subintervals shrinking to those points. The product-density conclusion is the Lebesgue differentiation theorem.  $\square$

**Lemma 7.9** (0/1 positive kernels). *Let  $U$  be a 0/1 graphon.*

- (i) *If  $Q_U(h) \geq 0$  for every bounded  $h$ , then  $U$  is a cluster graphon.*
- (ii) *If  $Q_U(h) \geq 0$  for every bounded mean-zero  $h$ , then  $U$  is a cluster graphon as well.*

*Thus, up to null sets,  $\Omega$  is partitioned into measurable cliques of  $U$ , with  $U = 0$  between different parts, together with a possible zero-dust set.*

*Proof.* First recall the measurable characterization. A 0/1 graphon is a cluster graphon if and only if its induced  $P_3$  density is zero; this is an immediate instance of the standard induced-density characterization of hereditary graphon classes [9]. For completeness, if the induced density is zero, then for almost every edge  $xy$  the rows  $U(x, \cdot)$  and  $U(y, \cdot)$  agree almost everywhere: a point on which they differ would form an induced  $P_3$  with  $x, y$ . Thus every nonzero row-equivalence class is

a clique, two distinct such classes are anticomplete, and the remaining zero-row points form zero dust.

Suppose now that the induced  $P_3$  density is positive. Apply [Lemma 7.8](#) to the set of triples  $(x, y, z)$  satisfying

$$U(x, y) = U(y, z) = 1, \quad U(x, z) = 0.$$

For arbitrarily small  $\xi > 0$  this gives equal-measure sets  $A, B, C$  for which the average of  $U$  is at least  $1 - \xi$  on  $A \times B$  and  $B \times C$ , and at most  $\xi$  on  $A \times C$ . Put  $f_S = \mathbf{1}_S/\mu(S)$ . Since every diagonal average is at most one,

$$Q_U(f_A - f_B + f_C) \leq 3 - 4(1 - \xi) + 2\xi < 0$$

for small  $\xi$ , contradicting (i). For (ii), the function  $f_A - 2f_B + f_C$  has mean zero and

$$Q_U(f_A - 2f_B + f_C) \leq 6 - 8(1 - \xi) + 2\xi < 0$$

for small  $\xi$ , again a contradiction.  $\square$

**Lemma 7.10** (Hypersimplex formula). *Suppose  $W$  is complete multipartite with positive-measure independent classes of sizes*

$$p_1 \geq p_2 \geq \dots$$

*and with the remaining mass forming clique dust. Then*

$$\text{til}(K_r, W) = \min_{0 \leq j \leq r-1} \frac{1 - \sum_{h=1}^j p_h}{r - j}. \quad (39)$$

*Proof.* Write  $d = 1 - \sum_h p_h$  for the dust mass. If a tiling has size  $q$ , its load in the  $h$ th independent class is at most  $\min\{p_h, q\}$ , whereas its total load is  $rq$ . Hence

$$d + \sum_{h \geq 1} \min\{p_h, q\} \geq rq. \quad (40)$$

Equivalently, for every  $0 \leq j \leq r - 1$  the total load outside the first  $j$  classes gives

$$rq \leq jq + 1 - \sum_{h=1}^j p_h,$$

which is the upper bound in (39). Conversely, if all these upper bounds hold, then (40) holds: if  $m = |\{h : p_h > q\}| < r$ , it is exactly the bound with  $j = m$ , while if  $m \geq r$  it is automatic.

It remains to realize the loads. The case  $q = 0$  is trivial, so assume  $q > 0$ . Choose numbers

$$0 \leq y_h \leq \min\{p_h, q\}, \quad 0 \leq y_D \leq d, \quad \sum_h y_h + y_D = rq;$$

they exist by (40). If  $d > 0$ , partition the atomless dust into  $N$  equal measurable pieces, where  $N$  is large enough that  $y_D/N \leq q$ , and prescribe load  $y_D/N$  on each piece. Together with the independent classes, these pieces form a countable complete support after we temporarily delete the edges within each dust piece. Normalize all prescribed loads by  $q$ . The resulting sequence  $z = (z_i)$  satisfies

$$0 \leq z_i \leq 1, \quad \sum_i z_i = r.$$

The countable hypersimplex is the closed convex hull of the incidence vectors of  $r$ -subsets; equivalently, every such vector is the vector of one-point marginals of a probability distribution on  $r$ -subsets. For completeness, choose  $N_0$  so that  $\sum_{i > N_0} z_i \leq 1$ , replace the tail by one coordinate of that mass, and apply [Lemma 5.1](#). If the tail mass is positive, then whenever the tail coordinate is selected, replace it by  $i > N_0$  with probability  $z_i / \sum_{j > N_0} z_j$ ; if it is zero, omit this step. This gives a probability distribution on  $r$ -subsets with one-point marginals  $z_i$ .

Multiplying that distribution by  $q$  gives a weighted fractional  $K_r$ -tiling whose prescribed load on every block is at most its block measure. Lifting each weighted clique uniformly to the corresponding ordered products of blocks gives a measurable countable nonnegative sum; on a block of mass  $a_i$  its pointwise load is  $y_i/a_i \leq 1$ . Consequently this is a graphon fractional tiling of size  $q$  in the edge-deleted step subgraphon, hence also in  $W$ . Thus every  $q$  satisfying the displayed upper bounds is feasible, proving (39).  $\square$

**Lemma 7.11** (Rank-two equality). *Let  $2/5 < y \leq 1/2$ . If  $\text{til}(K_2, W) \leq y$  and  $e(W) = 2y^2$ , then  $W$  is, up to weak isomorphism, a clique of measure  $2y$  together with an isolated set of measure  $1 - 2y$ .*

*Proof.* If  $y = 1/2$ , then  $e(W) = 1/2$  forces  $W = 1$  almost everywhere, which is the asserted terminal case. Hence assume  $y < 1/2$ . Use the graphon Erdős–Gallai inequality  $e(U) \leq F_2(\text{til}(K_2, U))$ . Since  $F_2$  is strictly increasing and its active branch at  $y > 2/5$  is  $2y^2$ , the hypotheses force  $\text{til}(K_2, W) = y$ . Let  $c$  be a minimum fractional vertex cover of mass  $y$ . Replacing  $W$  by its support shows, exactly as in Lemma 7.4, that  $W$  is 0/1-valued.

Let  $h$  be bounded and mean-zero, and replace the base measure by  $d\mu_t = (1 + th) d\mu$ . The same cover has mass  $y + t \int ch$ , while the edge density is  $2y^2 + t \int d_W h + O(t^2)$ . Since the clique branch remains active for small positive and negative  $t$ , and since the same  $c$  is a cover of the perturbed graphon with mass  $y + t \int ch$ , the graphon Erdős–Gallai inequality gives comparison with  $2(y + t \int ch)^2$ . The two signs of  $t$  therefore imply

$$\int d_W h = 4y \int ch.$$

Thus  $d_W - 4yc$  is constant almost everywhere. Its integral is zero, because

$$\int d_W = 2e(W) = 4y^2 = 4y \int c.$$

Therefore

$$d_W = 4yc. \tag{41}$$

In particular  $c \leq 1/(4y) < 1$  almost everywhere. Put  $P = \{c > 0\}$  and  $Z = \{c = 0\}$ . Let

$$E = \{(z, v) \in Z \times \Omega : W(z, v) = 1\}.$$

For almost every  $(z, v) \in E$ , the cover inequality gives  $c(v) \geq 1$ . On the other hand, (41) and  $d_W \leq 1$  give  $c(v) \leq 1/(4y) < 1$  almost everywhere. Hence  $\mu^2(E) = 0$ , so  $W = 0$  almost everywhere on  $Z \times \Omega$ . By Lemma 7.3(i), now with  $r = 2$ ,  $\mu(P) \leq 2y$ . Since  $W$  is supported on  $P^2$ ,

$$2y^2 = e(W) \leq \frac{\mu(P)^2}{2} \leq 2y^2.$$

Equality throughout gives  $\mu(P) = 2y$  and  $W = 1$  almost everywhere on  $P^2$ . Thus  $P$  is the required clique and  $Z$  is isolated.  $\square$

## 8. DIRECT VARIATIONAL CERTIFICATES IN AN OPEN PHASE

Fix an open phase on which a single quadratic function  $F$  equals  $F_r$ . Let  $W$  be extremal at  $x$  and let  $c$  be a minimum cover of mass  $x$ .

**Theorem 8.1** (Exact first and second variation). *Assume  $x$  lies in the interior of one phase and*

$$\text{til}(K_r, W) = x, \quad e(W) = F_r(x) = F(x).$$

*Then:*

- (i)  $W$  is 0/1-valued almost everywhere;
- (ii)

$$d_W(v) = 2F(x) - xF'(x) + F'(x)c(v) \quad \text{a.e.};$$

(iii) for every bounded  $h$  with  $\int h = 0$ ,

$$Q_W(h) \leq F''(x) \left( \int ch \right)^2.$$

*Proof.* Assertion (i) is [Lemma 7.4](#).

Let  $h$  be bounded with  $\int h = 0$ , and change the base measure to

$$d\mu_t = (1 + th) d\mu$$

for small positive and negative  $t$ . The same cover has new mass

$$x_t = x + t \int ch.$$

Writing

$$a = \int ch, \quad b = \int d_W h, \quad q = Q_W(h),$$

the edge density becomes

$$e_t = e(W) + tb + \frac{t^2}{2}q.$$

For both signs of  $t$  the parameter remains in the same phase. Although  $c$  need not remain minimum, it is still a cover of mass  $x_t$ ; hence the density theorem and monotonicity give

$$e_t \leq F_r(\text{fcov}(K_r, W, \mu_t)) \leq F_r(x_t) = F(x_t) = F(x) + tF'(x)a + \frac{t^2}{2}F''(x)a^2.$$

Equality at  $t = 0$  and the two signs imply  $b = F'(x)a$ , hence  $d_W - F'(x)c$  is constant. Integrating determines the constant. Comparing quadratic terms gives the last assertion.  $\square$

The three open-phase formulas are

$$B'_{r,i}(x) = \frac{i(1 - ix)}{r - i}, \quad B''_{r,i}(x) = -\frac{i^2}{r - i}, \quad (42)$$

$$A'_r(x) = 4x, \quad A''_r(x) = 4, \quad (43)$$

$$R'_r(x) = r(1 - (r - 1)(1 - rx)), \quad R''_r(x) = r^2(r - 1). \quad (44)$$

## 9. EXACT CLASSIFICATION IN A $B_{r,i}$ PHASE

Let  $s = r - i$  and  $a = 1 - ix$ . For an extremizer in the  $B_{r,i}$  phase, [Theorem 8.1](#) gives

$$d_W = 1 - \frac{a}{s} + \frac{ia}{s}c. \quad (45)$$

Put  $f = 1 - ic$ . Since  $c \geq 0$  and  $d_W \leq 1$ , one has  $0 \leq f \leq 1$ , and

$$d_{\overline{W}} = \frac{a}{s}f, \quad \int f = a.$$

For mean-zero  $h$ ,

$$Q_{\overline{W}}(h) = -Q_W(h) \geq \frac{1}{s} \left( \int fh \right)^2.$$

Thus

$$K = \overline{W} - \frac{1}{s}f \otimes f$$

satisfies  $K\mathbf{1} = 0$  and is positive semidefinite on the bounded mean-zero functions. By [Lemma 7.5](#), this extends to the closed mean-zero subspace of  $L^2$ . It is therefore positive semidefinite on all of  $L^2$ , and so is  $\overline{W}$ . By [Lemma 7.9](#),  $\overline{W}$  is a cluster graphon. Equivalently,  $W$  is complete multipartite with a possible clique-dust region.

Let the positive-measure independent classes have sizes  $p_1 \geq p_2 \geq \dots$ . Since  $W$  is extremal,

$$\sum_h p_h^2 = 1 - 2e(W) = \frac{a^2}{s}. \quad (46)$$

By Lemma 7.10 and  $\text{til}(K_r, W) = x < 1/r$ , the minimizing index satisfies  $1 \leq j \leq r-1$ . With  $q = r-j$ ,

$$\sum_{h=1}^j p_h = 1 - qx.$$

Hence

$$\sum_h p_h^2 \geq \frac{(1 - qx)^2}{r - q} = 1 - 2B_{r,q}(x). \quad (47)$$

If  $B_{r,i}$  is uniquely active, (46) and (47) force  $q = i$ . Equality in Cauchy-Schwarz then gives

$$p_1 = \dots = p_s = \frac{a}{s}, \quad p_h = 0 \quad (h > s).$$

The remaining mass is  $1 - a = ix$  and is clique dust. This is exactly  $W_{B_i}(x)$ .

**Theorem 9.1** (Open  $B$ -phase classification). *If  $B_{r,i}$  is the unique active curve at  $x$ , then*

$$\mathcal{M}_r(x) = \{W_{B_i}(x)\}.$$

The same argument will be used at transition endpoints.

**Lemma 9.2** ( $B$ -certificate at a transition). *Suppose  $e(W) = F_r(x)$ ,  $c$  is a minimum cover of mass  $x$ , and the degree and second-order identities associated with a  $B_{r,i}$  branch hold. Then  $W$  is a  $B_{r,q}$  candidate for some  $B_q \in \mathcal{A}_r(x)$ .*

*Proof.* The positive-semidefinite argument above still gives a complete multipartite graphon with deficiency (46). Equality in (47) forces  $B_{r,q}(x) = F_r(x)$  and forces exactly  $r - q$  equal independent classes of total mass  $1 - qx$ . The remaining mass is clique dust, which is the  $B_{r,q}$  construction.  $\square$

## 10. EXACT CLASSIFICATION IN THE $A_r$ PHASE

Let

$$\rho = \frac{1}{r-1}, \quad \lambda = \frac{r-2}{r-1}.$$

The  $A$ -certificate is

$$d_W = \lambda + 4xc, \quad Q_W(h) \leq 4 \left( \int ch \right)^2 \left( \int h = 0 \right). \quad (48)$$

Define

$$U = \overline{W} + 4c \otimes c.$$

Then  $U\mathbf{1} = \rho\mathbf{1}$ . The second inequality in (48) says that  $Q_U \geq 0$  on the bounded mean-zero functions. By Lemma 7.5, the inequality extends to the closed mean-zero subspace; the constant row sum then shows that  $U$  is positive semidefinite on all of  $L^2$ .

**10.1. The zero layer is non-null.** Put  $P = \{c > 0\}$  and  $Z = \{c = 0\}$ . The near-optimal tiling argument in Lemma 7.3(i) gives

$$\mu(P) \leq rx, \quad \mu(Z) \geq 1 - rx > 0. \quad (49)$$

No attained optimal tiling, pointwise load identity, or robust tight-clique assertion is needed in the  $A$  phase.

**10.2. The zero-layer components.** On  $Z^2$  one has  $U = \overline{W}$ . Hence [Lemma 7.9](#) shows that  $\overline{W}[Z]$  is a cluster graphon. Write its positive-measure clique components as

$$C_1, \dots, C_m.$$

There is no positive-measure zero dust: such a set would be a positive-measure clique of  $W$  with zero cover. Also  $m \leq r - 1$ , since one point from each of  $r$  components would form a zero-cover  $K_r$  in  $W$ .

For  $h$  supported on  $C_j$  with  $\int_{C_j} h = 0$ , one has  $Q_U(h) = 0$ . Positive semidefiniteness gives  $T_U h = 0$ . By [Lemma 7.7](#), there is a common global row  $g_j$  for almost all points of  $C_j$ . Since  $c = 0$  on  $C_j$ , this is a 0/1 row of  $\overline{W}$ . Put

$$S_j = \{y : g_j(y) = 1\}.$$

The constant row sum gives  $\mu(S_j) = \rho$ , and  $C_j \subseteq S_j$  modulo null sets.

Let  $p_j = \mu(C_j)$  and  $e_j = \mathbf{1}_{C_j} / \sqrt{p_j}$ . Then

$$T_U e_j = \sqrt{p_j} \mathbf{1}_{S_j}, \quad \langle e_j, T_U e_j \rangle = p_j, \quad \|T_U e_j\|_2^2 = \rho p_j.$$

The Schur test gives  $\|T_U\| \leq \rho$ . Since  $T_U \geq 0$ , the operator  $\rho T_U - T_U^2$  is positive semidefinite. Its quadratic form vanishes on  $e_j$ , hence

$$T_U \mathbf{1}_{S_j} = \rho \mathbf{1}_{S_j}.$$

As  $U \geq 0$  and every row has integral  $\rho$ , it follows that

$$U = 0 \quad \text{a.e. on } S_j \times (\Omega \setminus S_j). \quad (50)$$

The sets  $S_j$  are pairwise disjoint: indeed,

$$\langle T_U e_j, T_U e_k \rangle = \langle e_j, T_U^2 e_k \rangle = \rho \langle e_j, T_U e_k \rangle = 0$$

for  $j \neq k$ ; the last equality uses  $U = \overline{W} = 0$  on  $C_j \times C_k$ .

Since  $U \geq 4c \otimes c$ , positive cover mass cannot occur in two different  $U$ -components. Thus either  $P$  is contained in one of the  $S_j$ , or  $P$  is disjoint from every  $S_j$  and lies in the residual set  $\Omega \setminus \bigcup_j S_j$ .

**10.3. Reduction to rank two.** Assume first that  $P \subseteq S_1$ . Every zero-cover point belongs to one of the  $C_j \subseteq S_j$ , so no point lies outside  $\bigcup_j S_j$ . Therefore  $m\rho = 1$  and  $m = r - 1$ . For  $j \geq 2$ , the component  $S_j$  contains no positive-cover point and hence  $S_j = C_j$ . In the original graphon  $W$ , these  $r - 2$  sets are independent, complete to their complements, and have measure  $\rho$ .

Let  $W_1$  be the normalized restriction of  $W$  to  $S_1$ , and put  $y = (r - 1)x$ . The restriction of  $c$  is a fractional  $K_2$ -cover of  $W_1$  of mass  $y$ : every edge in  $W_1$ , together with one point from each of the  $r - 2$  peeled classes, forms a  $K_r$ . The complete joins between the  $r - 1$  components give

$$e(W) = \frac{r - 2}{2(r - 1)} + \rho^2 e(W_1).$$

Since  $e(W) = A_r(x)$ , one obtains  $e(W_1) = 2y^2$ . The phase diagram gives  $y > 2/5$ , and [Lemma 7.11](#) shows that  $W_1$  is a clique of measure  $2y$  plus an isolated remainder. Rescaling yields

$$\mu(X) = 2x, \quad \mu(Z \cap S_1) = \frac{1}{r - 1} - 2x,$$

so  $W \cong W_A(x)$ .

It remains to exclude the residual-component alternative. In that case every  $S_j = C_j$  is a zero-cover independent class complete to its complement. Peel all  $m$  classes and normalize the remainder. Put

$$t = r - m, \quad q = 1 - \frac{m}{r - 1} = \frac{t - 1}{r - 1}, \quad y = \frac{x}{q}.$$

The residual set contains  $P$  and therefore has positive measure, so  $t \geq 2$  and  $q > 0$ . The restricted cover is a minimum fractional  $K_t$ -cover. First, every  $K_t$  in the remainder extends to a  $K_r$  by choosing one point from each peeled class, so the restriction of  $c$  is a  $K_t$ -cover. Conversely, every  $K_r$  uses at most one point from each peeled independent class and therefore contains a  $K_t$  in the remainder. Thus any smaller  $K_t$ -cover of the remainder, extended by zero on the peeled classes, would cover every  $K_r$  and contradict the minimality of  $c$ . The residual cover is positive almost everywhere, so [Lemma 7.3\(i\)](#), applied after normalization, gives  $1 \leq ty$ . But

$$y < \frac{1}{2(t-1)} \leq \frac{1}{t},$$

where the first inequality is the strict upper bound of the  $A$  phase. This contradiction finishes the classification.

**Theorem 10.1** (Open  $A$ -phase classification). *If  $A_r$  is uniquely active at  $x$ , then*

$$\mathcal{M}_r(x) = \{W_A(x)\}.$$

**Lemma 10.2** ( $A$ -certificate at  $B_1 = A$ ). *Let  $x$  be the nonzero transition  $B_{r,1}(x) = A_r(x)$ , let  $W \in \mathcal{M}_r(x)$ , and let  $c$  be a minimum cover. If the degree and second-order relations in the  $A$ -certificate hold, then  $W \cong W_A(x)$ .*

*Proof.* The preceding proof only used  $2/(4r-3) \leq x < 1/[2(r-1)]$ , the  $A$ -certificate, and equality  $e(W) = A_r(x)$ . All three hold at this transition.  $\square$

## 11. EXACT CLASSIFICATION IN THE $R_r$ PHASE

Let

$$\ell = 1 - rx, \quad \alpha = 1 - (r-1)\ell.$$

The  $R$ -certificate is

$$d_W = (r-2)\ell + r\alpha c. \tag{51}$$

In the open  $R$  phase,

$$\ell < \frac{1}{r^2 - r + 1}, \tag{52}$$

which is equivalent to the positive gap

$$g_R = r^2x - (r-1 + (r-1)x) > 0. \tag{53}$$

Indeed, at  $\ell_0 = 1/(r^2 - r + 1)$  one has

$$(d_B - d_R)(\ell_0) = -\frac{r^2 - 3r + 3}{(r^2 - r + 1)^2} < 0,$$

whereas  $(d_B - d_R)(0) = 1/r^2 > 0$ . Since the effective crossing is the smaller root from [Lemma 2.3](#), it is strictly smaller than  $\ell_0$ . The open  $R$  phase is the interval of smaller  $\ell$ , which proves [\(52\)](#).

Put  $P = \{c > 0\}$  and  $Z = \{c = 0\}$ . As before, [Lemma 7.3](#) gives  $\mu(P) \leq rx$ .

**Lemma 11.1.** *Let  $x$  lie in the open  $R$  phase, let  $W \in \mathcal{M}_r(x)$ , and let  $c$  be a minimum cover. With the preceding notation, there is  $\delta > 0$  such that, up to a  $\mu^r$ -null set, no clique tuple of cover sum below  $1 + \delta$*

- (i) *meets  $Z$ ; or*
- (ii) *contains a point  $v$  with  $c(v) \geq 1/(r-1)$ .*

*Proof.* Choose  $0 < \delta < 1/[2(r-1)]$  so that

$$\frac{x}{1/(r-1) - \delta} < (r-1)x + \frac{g_R}{2}.$$

We first isolate the full-measure set of clique tuples on which the pointwise argument is legitimate. Delete tuples having a coordinate where (51) fails, and delete the null set on which the cover inequality fails. For each  $i \in [r]$ , apply Fubini to the set of  $(v_1, \dots, v_r, w)$  for which the tuple obtained by replacing  $v_i$  by  $w$  is a clique but violates the cover inequality. After deleting one further null set of  $Q = (v_1, \dots, v_r)$ , the following holds simultaneously for every  $i$ : for almost every  $w$ , whenever the replacement tuple is a clique, it satisfies the cover inequality. Call such clique tuples  $Q$  typical.

Suppose that the set of tuples in either (i) or (ii) has positive measure, and choose a typical one. Put  $M = \max_i c(v_i)$  and choose an index  $i$  attaining the maximum. If  $Q$  meets  $Z$ , then its at most  $r-1$  positive values have sum at least one, so  $M \geq 1/(r-1)$ ; this lower bound is part of the hypothesis in case (ii). If  $c(w) < M - \delta$  and  $w$  were adjacent to every vertex of  $Q \setminus \{v_i\}$ , then the replacement tuple would be a clique of cover sum less than one, contrary to typicality. Hence, for almost every such  $w$ , it misses at least one of those  $r-1$  vertices. Writing  $\deg_Q(w) = \sum_{j=1}^r W(v_j, w)$  and integrating gives

$$\sum_{i=1}^r d_W(v_i) = \int \deg_Q(w) \, dw \leq r-1 + \mu\{c \geq M - \delta\} \leq r-1 + \frac{x}{1/(r-1) - \delta} < r-1 + (r-1)x + \frac{g_R}{2}.$$

On the other hand, (51) and the cover inequality on the typical tuple give

$$\sum_{i=1}^r d_W(v_i) \geq r(r-2)\ell + r\alpha = r(1-\ell) = r^2x = r-1 + (r-1)x + g_R,$$

a contradiction.  $\square$

By Lemma 7.3(ii),(iv) and part (i) of Lemma 11.1, the load of a near-optimal tiling on  $Z$  tends to zero. Hence

$$rx = \lim_n \int \ell_n = \lim_n \int_P \ell_n \leq \mu(P) \leq rx.$$

Thus

$$\mu(P) = rx, \quad \mu(Z) = \ell. \tag{54}$$

Part (ii) and Lemma 7.3(v) show that

$$\mu\{c \geq 1/(r-1)\} = 0.$$

Let  $H = \{c \geq 1/(r-1)\}$ , so  $\mu(H) = 0$ . Apart from the null set on which the cover inequality fails, every clique tuple meeting  $Z$  has its remaining at most  $r-1$  positive cover values summing to at least one, and therefore has a coordinate in  $H$ . The union of the coordinate cylinders  $\{x_i \in H\}$  has  $\mu^r$ -measure zero. Hence the set of  $K_r$ -tuples meeting  $Z$  has measure zero.

Apply Theorem 7.2 with  $\mu(Z) = \ell$ . Equality at  $R_r(x)$  forces  $W$  to lie in the  $R_r$  family.

**Theorem 11.2** (Open  $R$ -phase classification). *If  $R_r$  is uniquely active at  $x$ , then*

$$\mathcal{M}_r(x) = \mathcal{E}_{r,R}(x).$$

**Lemma 11.3** ( $R$ -certificate at  $B_{r,r-1} = R$ ). *Let  $x$  be the transition  $B_{r,r-1}(x) = R_r(x)$ , let  $W \in \mathcal{M}_r(x)$ , and let  $c$  be a minimum cover. If the degree relation corresponding to the  $R$ -slope holds, then  $W \in \mathcal{E}_{r,R}(x)$ .*



*Proof.* At that transition the lower degree-sum expression in the preceding proof exceeds the upper one by  $(m_R - m_B)(1 - rx) > 0$ . Thus the proof of [Lemma 11.1](#) applies with this gap. It gives (54) and a null high-cover set; the coordinate-cylinder argument immediately following that lemma then shows that the clique tuples meeting  $Z$  form a null set. The extension-of-Turán equality theorem now applies.  $\square$

## 12. NONSMOOTH MULTIPLIERS AND TRANSITION POINTS

Let  $x_0$  be a first-order transition at which the left and right active quadratics are  $f_-$  and  $f_+$ , with

$$m_- = f'_-(x_0) < f'_+(x_0) = m_+.$$

**Lemma 12.1** (Nonsmooth Euler-Lagrange multiplier). *Let  $W \in \mathcal{M}_r(x_0)$  and let  $c$  be a minimum cover. There is  $\theta \in [m_-, m_+]$  such that*

$$d_W = \lambda + \theta c, \quad \lambda = 2F_r(x_0) - \theta x_0. \quad (55)$$

If  $\theta = m_-$ , then

$$Q_W(h) \leq f''_-(x_0) \left( \int ch \right)^2$$

for every bounded mean-zero  $h$ ; the analogous statement holds with  $f_+$  if  $\theta = m_+$ . If  $m_- < \theta < m_+$ , then

$$Q_W(h) \leq 0 \quad \text{whenever} \quad \int h = \int ch = 0. \quad (56)$$

*Proof.* For mean-zero  $h$ , put  $a = \int ch$  and  $b = \int d_W h$ . If  $a = 0$ , both signs of a small base-measure perturbation keep the cover mass equal to  $x_0$ , so  $b = 0$ . Unless  $a$  is the zero functional on the mean-zero subspace, it follows that  $b = \theta a$  for a scalar  $\theta$ . In the exceptional case  $c$  is constant almost everywhere; then  $b = 0$  for every mean-zero  $h$ , and the same relation holds after choosing any scalar  $\theta$ . The two one-sided derivatives of  $F_r$  allow the choice  $m_- \leq \theta \leq m_+$ . Integrating determines  $\lambda$ .

If  $\theta$  is an endpoint, perturb in the direction entering the corresponding phase; the linear term cancels and the second-order comparison gives the asserted quadratic inequality. If  $\theta$  is strict and  $a = 0$ , both perturbation signs keep the parameter fixed, and comparison with  $F_r(x_0)$  gives (56).  $\square$

**12.1. The tangential transition**  $A_r = B_{r,2}$ . At  $x = 1/[2(r-1)]$  the two candidates coincide and their slopes agree; write the common slope as  $m$ . The first-order argument from [Theorem 8.1](#) remains two-sided at a differentiability point of  $F_r$  and gives

$$d_W = 2F_r(x) - mx + mc.$$

Let  $h$  be bounded and mean-zero, and put  $a = \int ch$ . If  $a \neq 0$ , choose the sign of  $t$  for which  $x + ta$  lies on the  $B_2$  side; if  $a = 0$ , either sign may be used. The linear term cancels, and the one-sided quadratic comparison gives

$$Q_W(h) \leq B''_{r,2}(x)a^2.$$

Thus the full  $B_2$  certificate holds. By [Lemma 9.2](#), the extremizer is the common degeneration  $W_A = W_{B_2}$ .

**12.2. The transition**  $B_{r,r-1} = R_r$ . Let  $x_*$  be the effective crossing, and write  $m_B < m_R$  for its slopes. If  $\theta = m_B$ , [Lemma 9.2](#) gives  $W_{B_{r-1}}(x_*)$ .

Assume  $\theta > m_B$ . Put  $Z = \{c = 0\}$ . At cover sum one, the degree relation (55) gives

$$2rF_r(x_*) + \theta(1 - rx_*).$$

For  $\theta = m_B$  this equals  $r - 1 + (r - 1)x_*$ . Thus for  $\theta > m_B$  there is a positive gap. Repeating the Fubini-typical argument of [Lemma 11.1](#), with this degree sum in place of  $r^2x$ , shows that for some  $\delta > 0$ , up to a null set, no clique tuple of cover sum below  $1 + \delta$  meets  $Z$  or contains a point of cover value at least  $1/(r - 1)$ . By [Lemma 7.3](#),

$$\mu(Z) = 1 - rx_*, \quad \mu\{c \geq 1/(r - 1)\} = 0.$$

Let  $H = \{c \geq 1/(r - 1)\}$ , so  $\mu(H) = 0$ . Outside the cover-exceptional null set, every clique tuple meeting  $Z$  has one of its remaining at most  $r - 1$  coordinates in  $H$ . The union of the corresponding coordinate cylinders is null, so the clique tuples meeting  $Z$  form a null set. Equality in [Theorem 7.2](#) gives an  $R_r$  graphon. Hence

$$\mathcal{M}_r(x_*) = \mathcal{E}_{r,B_{r-1}}(x_*) \cup \mathcal{E}_{r,R}(x_*). \quad (57)$$

### 13. THE DEFECT SYSTEM AT A FIRST-ORDER TRANSITION

The remaining first-order transitions are  $B_{r,1} = A_r$  and  $B_{r,i} = B_{r,i+1}$  for  $2 \leq i \leq r - 2$ . Assume for contradiction that the multiplier in [Lemma 12.1](#) is strict. Put

$$D = 1 - 2F_r(x), \quad L = D + \theta x, \quad d_W = 1 - L + \theta c. \quad (58)$$

Let  $Z = \{c = 0\}$  and  $P = \{c > 0\}$ . All endpoint slopes at these transitions are positive, so  $\theta > 0$ . By [Lemma 7.5](#), the inequality (56), initially proved for bounded functions, extends to the closed tangent subspace

$$\mathcal{H}_c = \left\{ h \in L^2 : \int h = \int ch = 0 \right\}.$$

By [Lemma 7.3](#),

$$z := \mu(Z) \geq \ell := 1 - rx. \quad (59)$$

**13.1. The zero layer.** For mean-zero  $h$  supported on  $Z$ , (56) gives  $Q_{\overline{W}}(h) \geq 0$ . By [Lemma 7.9\(ii\)](#),  $\overline{W}[Z]$  is a cluster graphon. There is no positive-measure zero dust, and therefore

$$Z = C_1 \sqcup \cdots \sqcup C_m, \quad m \leq r - 1,$$

where every  $C_j$  is independent in  $W$  and distinct  $C_j, C_k$  are completely joined.

If  $h$  is supported on  $C_j$  and has mean zero, then  $Q_W(h) = 0$ . By [Lemma 7.6](#),  $T_W h \in \text{span}\{\mathbf{1}, c\}$ . The function  $T_W h$  vanishes on  $Z$ : the  $W$ -row into  $C_j$  is constant on every zero-layer class. Hence the constant coefficient is zero. Also

$$\int T_W h = \int d_W h = 0$$

by (58), so the coefficient of  $c$  is zero. Thus  $T_W h = 0$ . By [Lemma 7.7](#),  $C_j$  is a global twin class.

Put  $p_j = \mu(C_j)$  and define the measurable normalized row average

$$W(y, C_j) = \frac{1}{p_j} \int_{C_j} W(y, z) \, dz.$$

The row-constancy lemma and the fact that  $W$  is 0/1-valued show that  $W(y, C_j) \in \{0, 1\}$  for almost every  $y$ . Put

$$D_j = \{y \in P : W(y, C_j) = 0\}, \quad q_j = \mu(D_j).$$

The degree of a point in  $C_j$  gives

$$p_j + q_j = L. \quad (60)$$

For  $y \in P$ , let

$$M(y) = \{j : y \in D_j\}, \quad \bar{d}_P(y) = \int_P (1 - W(y, z)) \, dz.$$

The pointwise missing-degree identity is

$$\theta c(y) = L - \sum_{j \in M(y)} p_j - \bar{d}_P(y). \quad (61)$$

Finally, with  $\Delta_P = \int_{P^2} (1 - W)$ ,

$$D = Lz + \sum_j p_j q_j + \Delta_P = mL^2 - \sum_j q_j^2 + \Delta_P. \quad (62)$$

**13.2. Crossing flat directions and finite uncrossing.** The next lemma is entirely integral; it does not require pointwise localization of rows or a false-twin argument.

**Lemma 13.1** (A flat crossing direction). *Let  $j \neq k$ , and suppose that*

$$A = D_j \setminus D_k, \quad B = D_k \setminus D_j$$

*have positive measure. Put*

$$f_S = \frac{\mathbf{1}_S}{\mu(S)}, \quad \alpha = \int c f_A, \quad \beta = \int c f_B,$$

*and define*

$$h = \beta(f_A - f_{C_j}) + \alpha(f_{C_k} - f_B). \quad (63)$$

*Then*

$$W = 1 \quad \text{a.e. on } A \times B, \quad W = 0 \quad \text{a.e. on } A^2 \cup B^2, \quad (64)$$

*and*

$$\int h = \int ch = 0, \quad T_W h = 0. \quad (65)$$

*Proof.* Because  $A, B \subseteq P$ , one has  $\alpha, \beta > 0$ . By the definitions of the defect sets and the global-twin property of the zero classes,

$$W(A, C_j) = W(B, C_k) = 0, \quad W(A, C_k) = W(B, C_j) = W(C_j, C_k) = 1$$

almost everywhere, while  $W = 0$  on  $C_j^2 \cup C_k^2$ . Also

$$\int h = 0, \quad \int ch = \beta\alpha - \alpha\beta = 0.$$

Write

$$w_{AB} = \int_{A \times B} W f_A f_B, \quad w_{AA} = Q_W(f_A), \quad w_{BB} = Q_W(f_B).$$

An exact expansion around the four classes gives

$$Q_W(h) = \beta^2 w_{AA} + \alpha^2 w_{BB} + 2\alpha\beta(1 - w_{AB}). \quad (66)$$

Every term on the right is nonnegative. On the other hand,  $h$  belongs to the tangent space in (56), so  $Q_W(h) \leq 0$ . Hence all three terms vanish. Since  $W$  is 0/1-valued, this proves (64) and also  $Q_W(h) = 0$ .

By Lemma 7.6,  $T_W h \in \text{span}\{\mathbf{1}, c\}$ . For a typical point of  $C_j$ , the contributions from  $C_k$  and  $B$  cancel and the other two contributions vanish; hence  $T_W h = 0$  on  $C_j$ . Similarly the contributions from  $A$  and  $C_j$  cancel on  $C_k$ , so  $T_W h = 0$  there as well. Since  $c = 0$  on both positive-measure classes, the constant coefficient in  $T_W h = a_0 \mathbf{1} + b_0 c$  is zero. Finally,

$$b_0 x = \int T_W h = \int d_W h = (1 - L) \int h + \theta \int ch = 0.$$

All transitions considered here have  $x > 0$ , so  $b_0 = 0$  and (65) follows.  $\square$

**Lemma 13.2** (Finite exact uncrossing). *Starting from any extremizer with a strict interior multiplier, finitely many changes of the base measure produce another extremizer with the same  $x$ , the same cover  $c$ , the same edge density, and the same degree relation (58), for which the defect sets are linearly ordered modulo null sets. After relabeling,*

$$D_1 \subseteq D_2 \subseteq \cdots \subseteq D_m. \quad (67)$$

*Proof.* Suppose  $D_j, D_k$  cross and take  $h$  from Lemma 13.1. Let

$$I = \{t \in \mathbb{R} : 1 + th \geq 0 \text{ a.e.}\} = [t_-, t_+].$$

The function  $h$  is positive on  $A \cup C_k$ , negative on  $C_j \cup B$ , and zero elsewhere, so  $t_- < 0 < t_+$  and at either endpoint at least one of the four sets  $A, B, C_j, C_k$  receives zero mass. Fix one endpoint  $t \in \{t_-, t_+\}$  and put

$$d\mu_t = (1 + th) d\mu.$$

This is again an atomless probability measure because it is absolutely continuous with respect to the atomless measure  $\mu$  and  $\int h = 0$ . The old cover inequalities hold  $\mu_t^t$ -almost everywhere, and

$$\int c d\mu_t = x + t \int ch = x.$$

Moreover, by (65),

$$d_{W, \mu_t} = d_{W, \mu} + t T_W h = d_{W, \mu},$$

and

$$e(W, \mu_t) = e(W, \mu) + t \int d_W h + \frac{t^2}{2} Q_W(h) = F_r(x).$$

Thus  $\text{fcov}(K_r, W, \mu_t) \leq x$ . The density theorem and the strict monotonicity of  $F_r$  give

$$F_r(x) = e(W, \mu_t) \leq F_r(\text{fcov}(K_r, W, \mu_t)) \leq F_r(x),$$

so the cover remains minimum and the new graphon remains extremal at  $x$ . The displayed degree identity shows that the same  $L$  and  $\theta$  remain valid. In particular, for every bounded  $g$  satisfying

$$\int g d\mu_t = \int cg d\mu_t = 0,$$

a further two-sided base-measure perturbation gives  $Q_{W, \mu_t}(g) \leq 0$ . Hence the whole strict-multiplier setup is inherited by the new measure.

Let  $\mathcal{P}_{\text{pat}}$  be the collection of positive-measure atoms of the Boolean algebra generated by  $D_1, \dots, D_m$  inside  $P$ , and define

$$\kappa = m + |\mathcal{P}_{\text{pat}}|.$$

If the endpoint move kills  $C_j$  or  $C_k$ , then  $m$  decreases; deleting that defect coordinate can only merge old membership patterns. If it kills  $A$  or  $B$ , at least one positive membership atom disappears. No new positive-measure pattern can appear because  $\mu_t \ll \mu$ . Consequently  $\kappa$  strictly decreases after every endpoint move. Since  $\kappa$  is a nonnegative integer, the process terminates. At termination no two defect sets cross on sets of positive measure, so they are pairwise comparable modulo null sets and may be relabeled as in (67).  $\square$

**13.3. The smallest defect class and the local recursion.** Apply Lemma 13.2 and relabel the zero classes so that (67) holds. Put

$$C = C_1, \quad D_0 = D_1, \quad p = \mu(C) = L - q, \quad q = \mu(D_0), \quad a = \int_{D_0} c, \quad z = \mu(Z).$$

By nesting, every point of  $D_0$  misses every zero-layer class. Define

$$I = \int_{D_0^2} (1 - W), \quad \beta = \int_{D_0 \times (\Omega \setminus (C \cup D_0))} (1 - W).$$

Integrating (61) over  $D_0$  gives

$$q^2 - \theta a = \beta + I. \quad (68)$$

Since  $D_0$  is anticomplete to  $Z \setminus C$ ,

$$\beta \geq q(z - p). \quad (69)$$

Let  $R_0 = \Omega \setminus (C \cup D_0)$  and  $\lambda = \mu(R_0) = 1 - L$ . The class  $C$  is complete to  $R_0$ . Every  $K_{r-1}$  in  $R_0$ , together with one point of  $C$ , is therefore a  $K_r$ ; since  $c = 0$  on  $C$ , the restriction of  $c$  to  $R_0$  is a fractional  $K_{r-1}$ -cover. After normalizing the measure on  $R_0$ , its mass is

$$y = \frac{x - a}{\lambda}.$$

At a  $B_{r,j}$  endpoint the peel size is

$$L_j = 1 - 2B_{r,j}(x) + B'_{r,j}(x)x = \frac{1 - jx}{r - j},$$

and

$$1 - (r - 1)x - L_j = \frac{(r - j - 1)(1 - rx)}{r - j} \geq 0.$$

At the  $A$  endpoint of  $B_1 = A$  one has  $L_A = 1/(r - 1)$ , and direct substitution of  $x = 2/(4r - 3)$  gives  $L_A < 1 - (r - 1)x$ . Since  $L$  lies between the two endpoint values,  $L \leq 1 - (r - 1)x$ . Consequently  $0 \leq y \leq 1/(r - 1)$ . Decomposing the edge density and using (68) gives the exact identity

$$e(W) = L(1 - L) + \lambda^2 e(W[R_0]/\lambda) + \frac{\theta a}{2} - \frac{\beta}{2}. \quad (70)$$

The density theorem for  $K_{r-1}$  therefore yields the local recursion

$$F_r(x) \leq L(1 - L) + \lambda^2 F_{r-1}\left(\frac{x - a}{\lambda}\right) + \frac{\theta a}{2} - \frac{\beta}{2}. \quad (71)$$

Combining (68), (69), and  $z \geq \ell$  gives

$$\theta a \leq q(L - z).$$

If  $L \geq \ell$  and  $L - z \leq 0$ , this already forces  $a = 0$  and  $q(L - z) = 0$ ; if  $L - z > 0$ , use  $q \leq L$  and  $L - z \leq L - \ell$ . In both cases,

$$\theta a \leq q(L - z) \leq L(L - \ell). \quad (72)$$

If  $\ell > L$ , the same inequalities force  $q = 0$ . Then  $D_0$  is null,  $C$  has measure  $L$  and is an independent class complete to its complement, and

$$F_r(x) \leq L(1 - L) + (1 - L)^2 F_{r-1}\left(\frac{x}{1 - L}\right). \quad (73)$$

**Lemma 13.3** (Strict Bellman equality set). *At the first-order transition  $B_{r,1} = A_r$  with  $r \geq 4$ , and at the effective transitions  $B_{r,i} = B_{r,i+1}$  with  $2 \leq i \leq r - 3$ , the right-hand side of (73) is at most  $F_r(x)$ . Equality holds only when  $L$  is one of the two endpoint peel sizes*

$$L_- = 1 - 2F_r(x) + m_-x, \quad L_+ = 1 - 2F_r(x) + m_+x.$$

*Proof.* Set  $u = 1 - L$ . Then (73) is exactly the inequality in Lemma 3.3, with  $q = u$ . The equality clause of that lemma leaves precisely the two maximizing values associated with the active branches. Since

$$1 - q_j^* = \frac{1 - jx}{r - j} = 1 - 2B_{r,j}(x) + B'_{r,j}(x)x, \quad 1 - \frac{r - 2}{r - 1} = 1 - 2A_r(x) + A'_r(x)x,$$

these two values of  $u$  are equivalent to  $L \in \{L_-, L_+\}$ .  $\square$

## 14. ELIMINATING STRICT INTERIOR MULTIPLIERS

14.1. **The transition**  $B_{r,1} = A_r$  **for**  $r \geq 4$ . Here

$$x = \frac{2}{4r-3}, \quad \ell = \frac{2r-3}{4r-3}.$$

The larger endpoint peel size is  $L_A = 1/(r-1)$ , and

$$\ell - L_A = \frac{2r^2 - 9r + 6}{(4r-3)(r-1)} > 0 \quad (r \geq 4).$$

Thus  $\ell > L$  for every strict interior multiplier. The universal peel (73) and Lemma 13.3 give a strict contradiction.

14.2. **The transition**  $B_{r,i} = B_{r,i+1}$  **with**  $2 \leq i \leq r-3$ . Put  $s = r-i \geq 3$ . At the crossing there is  $t > 0$  such that

$$1 - ix = t\sqrt{s}, \quad 1 - (i+1)x = t\sqrt{s-1}.$$

The endpoint peel sizes and the zero-set lower bound are

$$L_i = \frac{t}{\sqrt{s}}, \quad L_{i+1} = \frac{t}{\sqrt{s-1}},$$

$$\ell = t(s\sqrt{s-1} - (s-1)\sqrt{s}).$$

Moreover

$$\frac{\ell}{L_{i+1}} = \frac{(s-1)\sqrt{s}}{\sqrt{s} + \sqrt{s-1}} > 1.$$

Again  $\ell > L$  throughout the strict multiplier interval, and Lemma 13.3 gives a contradiction.

14.3. **The last adjacent transition for**  $r \geq 5$ . It remains to treat  $B_{r,r-2} = B_{r,r-1}$ . Put  $d = r-2 \geq 3$ ,  $\omega = \sqrt{2}-1$ , and

$$b = \frac{1}{1 + (d+1)\omega}.$$

At the crossing,

$$x = \omega b, \quad D = b^2, \quad \ell = (2 - \sqrt{2})b,$$

and the endpoint peel sizes are

$$L_0 = \frac{b}{\sqrt{2}}, \quad L_1 = b.$$

For  $L_0 < L < L_1$ , put  $\theta = (L - b^2)/x$  and  $\lambda = 1 - L$ . From (72),

$$0 \leq a \leq A_* := \frac{L(L - \ell)}{\theta}.$$

The relevant part of  $F_{r-1}$  is

$$\max\{P_d(y), R_{r-1}(y)\}, \quad P_d(y) = dy - \frac{d^2 y^2}{2} = B_{r-1,r-2}(y).$$

The exact calculations in Section B show, for every  $0 \leq a \leq A_*$ ,

$$L(1 - L) + \lambda^2 F_{r-1}\left(\frac{x-a}{\lambda}\right) + \frac{\theta a}{2} < F_r(x),$$

contradicting (71) even after the nonpositive term  $-\beta/2$  is discarded.

**14.4. The exceptional transitions  $r = 3$  and  $r = 4$ .** At  $r = 3$ , the transition  $B_{3,1} = A_3$  occurs at  $x = 2/9$ . The zero layer has at most two classes; the two-class case contradicts  $L < 1/2$ , and the one-class case reduces through (71) to the exact rank-two calculation in Section A.

At  $r = 4$ , the transition  $B_{4,2} = B_{4,3}$  occurs at  $x = (4 - \sqrt{2})/14$ . Here the reduced envelope  $F_3$  passes through its  $A_3$ ,  $B_{3,2}$ , and  $R_3$  branches. The complete piecewise calculation in Section C is strictly below  $F_4(x)$  for every interior multiplier.

*Proof of Theorem 1.3.* The inclusion  $\mathcal{E}_r^*(x) \subseteq \mathcal{M}_r(x)$  is Lemma 2.1. It remains to classify an arbitrary  $W \in \mathcal{M}_r(x)$ .

At  $x = 0$ , one has  $\text{til}(K_r, W) = 0$ , so the set of clique tuples meeting  $Z = \Omega$  has measure zero. Applying Theorem 7.2 with  $z = 1$  gives the balanced complete  $(r - 1)$ -partite graphon; the  $B_1$  and  $A$  descriptions coincide with this graphon. At  $x = 1/r$ , equality  $e(W) = 1/2$  forces the complete graphon, the terminal member of the  $R$  family.

On open phases, apply Theorems 9.1, 10.1 and 11.2. At  $A = B_2$  use the tangential argument, and at  $B_{r,r-1} = R$  use (57). At every remaining first-order transition, an endpoint multiplier gives an active candidate by Lemmas 9.2, 10.2 and 11.3; the preceding elimination rules out a strict interior multiplier. The phase diagram in Lemma 2.3 exhausts all parameters.  $\square$

## 15. FORMAL STABILITY CONSEQUENCES OF EXACT CLASSIFICATION

This section uses only the classification (3), not the details of its proof.

### 15.1. Cut distance to a finite 0/1 template implies $L^1$ distance.

**Lemma 15.1.** *Let  $U$  be a 0/1 step graphon with a partition into at most  $M$  blocks. Then*

$$\delta_1(W, U) \leq M^2 \delta_\square(W, U).$$

*Proof.* Fix  $\zeta > 0$  and choose a relabeling for which the cut norm is at most  $\delta_\square(W, U) + \zeta$ . Let  $P_1, \dots, P_m$ ,  $m \leq M$ , be the blocks of  $U$ . On each rectangle  $P_a \times P_b$ , the function  $W - U$  has a fixed sign because  $U$  is identically 0 or 1 there. Hence

$$\begin{aligned} \|W - U\|_1 &= \sum_{a,b} \int_{P_a \times P_b} |W - U| \\ &= \sum_{a,b} \left| \int_{P_a \times P_b} (W - U) \right| \\ &\leq m^2 \|W - U\|_\square \leq M^2 (\delta_\square(W, U) + \zeta). \end{aligned}$$

Let  $\zeta \downarrow 0$ .  $\square$

### 15.2. Continuity and compactness of candidate families.

**Lemma 15.2.** *For fixed  $r$ , candidate families have the following properties.*

- (i) *Every candidate has at most  $2r - 1$  blocks.*
- (ii) *If  $x_n \rightarrow x$  and  $U_n \in \mathcal{E}_{r,\tau}(x_n)$ , a subsequence converges in  $L^1$  to a member of  $\mathcal{E}_{r,\tau}(x)$  or to the adjacent degenerate candidate at a domain endpoint.*
- (iii) *If  $U \in \mathcal{E}_r^*(x)$  and  $x_n \rightarrow x$ , there are  $\eta_n \rightarrow 0$  and  $U_n \in \mathcal{E}_r(x_n; \eta_n)$  such that  $\delta_1(U_n, U) \rightarrow 0$ .*

*Proof.* For  $B_i$  and  $A$ , all block measures are continuous affine functions of  $x$ . At  $x = 1/[2(r - 1)]$ , the  $Z$ -block of the  $A$  graphon disappears and the limit is the  $B_2$  graphon. For  $R$ , the parameter simplex

$$\{(z_1, \dots, z_{r-1}) \in [0, \ell]^{r-1} : \sum_j z_j = \ell\}$$

is compact and depends continuously on  $\ell = 1 - rx$ . Scaling the  $z_j$  gives the required approximants when  $\ell > 0$ ; when  $\ell \rightarrow 0$ , all members converge to the complete graphon. At a transition point, an active type on one side is automatically near-active on the other side because the candidate curves are continuous.  $\square$

### 15.3. Uniform graphon stability.

*Proof of Theorem 1.4 from (3).* Suppose the theorem fails. Then there are fixed  $\varepsilon, \eta > 0$ , parameters  $x_n$ , and graphons  $W_n$  such that

$$\text{til}(K_r, W_n) \leq x_n, \quad e(W_n) \geq F_r(x_n) - 1/n,$$

but  $\delta_1(W_n, \mathcal{E}_r(x_n; \eta)) \geq \varepsilon$ .

Take a subsequence with  $x_n \rightarrow x$  and  $W_n \rightarrow W$  in cut distance. The density theorem gives  $e(W_n) \leq F_r(x_n)$ , so  $e(W_n) \rightarrow F_r(x)$ . Lower semicontinuity gives  $\text{til}(K_r, W) \leq x$ . Hence  $e(W) = F_r(x)$  and, by (3),  $W \in \mathcal{E}_r^*(x)$ .

By Lemma 15.2, choose  $U_n \in \mathcal{E}_r(x_n; \eta_n)$ ,  $\eta_n \rightarrow 0$ , with  $\delta_1(U_n, W) \rightarrow 0$ . Then  $\delta_{\square}(W_n, U_n) \rightarrow 0$ . Since each  $U_n$  has at most  $2r - 1$  blocks, Lemma 15.1 yields  $\delta_1(W_n, U_n) \rightarrow 0$ . For large  $n$ ,  $U_n \in \mathcal{E}_r(x_n; \eta)$ , a contradiction.  $\square$

### 15.4. Rounding to finite graphs.

**Lemma 15.3** (Finite-template rounding). *Let  $U_n$  be 0/1 step graphons with at most  $M$  blocks, and let  $G_n$  be  $n$ -vertex graphs. If  $\delta_1(W_{G_n}, U_n) = o(1)$ , then there are finite blow-ups  $H_n$  of the templates  $U_n$  such that*

$$\text{edit}(G_n, H_n) = o(n^2).$$

*If  $U_n$  is a candidate graphon at  $x_n = k_n/n$ , then  $H_n$  can be chosen as an integer rounding of the corresponding candidate construction.*

*Proof.* Relabel  $U_n$ , while keeping the standard vertex-interval representation of  $W_{G_n}$  fixed, so that  $\|W_{G_n} - U_n\|_1 = o(1)$ . Let  $P_1, \dots, P_m$  be the blocks of  $U_n$ . A vertex interval  $I_v$  may intersect several  $P_j$ . Put

$$\alpha_{v,j} = n\mu(I_v \cap P_j), \quad \sum_j \alpha_{v,j} = 1.$$

Independently assign  $v$  to part  $j$  with probability  $\alpha_{v,j}$ , and then impose the 0/1 template. The expected number of incorrect unordered pairs is

$$\frac{n^2}{2} \|W_{G_n} - U_n\|_1 + O(n) = o(n^2).$$

Write this expectation as  $\eta_n n^2$ , where  $\eta_n \rightarrow 0$ . Markov's inequality shows that the number of incorrect pairs is at most  $\sqrt{\eta_n} n^2$  with probability  $1 - o(1)$ . Since the number of parts is bounded, Hoeffding's inequality and a union bound show that all part sizes differ from their expectations by  $O_M(\sqrt{n})$  with probability bounded away from zero. The two events therefore intersect for all large  $n$ . Choose an assignment in their intersection and impose the template. The resulting graph  $H_n$  proves the first assertion.

If  $U_n$  is a candidate at  $x_n = k_n/n$ , the integer construction in Section 2.5 has prescribed part sizes differing from the corresponding block expectations by  $O_r(1)$ . Starting with the same assignment, move  $O_M(\sqrt{n}) + O_r(1)$  vertices to reach those sizes. This costs  $O_{M,r}(n^{3/2}) = o(n^2)$  further edits and proves the second assertion.  $\square$

*Proof of Theorem 1.5 from (3).* If the theorem fails, there are fixed  $\varepsilon, \eta > 0$ , integers  $n_j \rightarrow \infty$  and  $0 \leq k_j \leq \lfloor n_j/r \rfloor$ , and  $n_j$ -vertex counterexamples  $G_j$  such that

$$\nu(K_r, G_j) \leq k_j, \quad e(G_j) \geq \left(F_r(k_j/n_j) - \frac{1}{j}\right) n_j^2,$$



and whose edit distance from every  $\eta$ -active candidate is at least  $\varepsilon n_j^2$ . Pass to a subsequence with  $k_j/n_j \rightarrow x$  and  $W_{G_j} \rightarrow W$  in cut distance. The finite-to-graphon transference inequality (35) gives

$$\text{til}(K_r, W) \leq \liminf_j \frac{\nu(K_r, G_j)}{n_j} \leq x.$$

Edge convergence and the density theorem yield  $e(W) = F_r(x)$ . By (3),  $W$  is an active candidate at  $x$ . Use Lemma 15.2 to choose  $\eta_j \rightarrow 0$  and  $U_j \in \mathcal{E}_r(k_j/n_j; \eta_j)$  with  $\delta_1(U_j, W) \rightarrow 0$ . After discarding finitely many terms,  $\eta_j < \eta$ . Then  $\delta_{\square}(W_{G_j}, U_j) \rightarrow 0$ , hence  $\delta_1(W_{G_j}, U_j) \rightarrow 0$  by Lemma 15.1. Apply Lemma 15.3, contradicting the edit-distance assumption.  $\square$

*Proof of Corollaries 1.6 and 1.7.* For each integer  $j \geq 1$ , apply Theorem 1.5 with  $\varepsilon = \eta = 1/j$ , obtaining parameters  $\delta_j, n_j$ . Choose a nondecreasing sequence  $j(n) \rightarrow \infty$  so slowly that  $n \geq n_{j(n)}$  and the edge deficit of  $G_n$  is at most  $\delta_{j(n)} n^2$ . The candidate supplied by the theorem then satisfies

$$\frac{\text{edit}(G_n, E_n)}{n^2} \leq \frac{1}{j(n)}, \quad F_r(x_n) - f_{r, \tau_n}(x_n) \leq \frac{1}{j(n)},$$

which proves the sequence form.

For the phasewise statement, for each  $\tau \neq \tau_0$  restrict to the compact set  $J$  intersected with the displayed domain of  $f_{r, \tau}$ , omitting empty intersections. On every such set the continuous gap  $F_r - f_{r, \tau}$  is strictly positive. Since there are finitely many types, the minimum of these positive minima is a number  $\gamma_J > 0$ . Apply Theorem 1.5 with any  $\eta < \gamma_J$ ; then the selected type must be  $\tau_0$ .  $\square$

#### DECLARATION ON THE USE OF AI

The authors used generative AI tools to assist in discussing proof strategies, checking proofs, and improving exposition.

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#### APPENDIX A. THE $r = 3$ TRANSITION $B_{3,1} = A_3$

At  $x = 2/9$ ,

$$F_3(x) = \frac{113}{324}, \quad D = \frac{49}{162},$$

and the endpoint slopes are  $7/18$  and  $8/9$ . Let a strict interior multiplier be given and put

$$L = D + \theta x, \quad \frac{7}{18} < L < \frac{1}{2}.$$

Here

$$\frac{\theta}{L} = \frac{9u + 14}{2u + 14} > 1$$

once  $u = 18L - 7 > 0$  is introduced below. Thus the degree bound  $1 - L + \theta c \leq 1$  gives  $c < 1$ , so every triangle contains at least two positive-cover points. The zero layer has at most two complete-multipartite parts.

If it has two parts, every positive point must miss at least one part; otherwise it forms a triangle with one zero point from each part. If  $q_1, q_2$  are the two defect measures, then

$$q_1 + q_2 \geq \mu(P).$$

Using  $p_j + q_j = L$  and  $\mu(P) + \mu(Z) = 1$  gives  $2L \geq 1$ , a contradiction.

Thus the zero layer has one part  $C$ , of measure  $p \geq 1 - 3x = 1/3$ . Put  $q = L - p$ ,  $a = \int_{D_0} c$ . Then

$$0 \leq a \leq A = \frac{(L - 1/3)^2}{\theta}.$$

The local rank-two recursion requires

$$\frac{113}{324} \leq \Psi(a) := L(1 - L) + (1 - L)^2 F_2\left(\frac{2/9 - a}{1 - L}\right) + \frac{\theta a}{2}.$$

Set  $u = 18L - 7 \in (0, 2)$ . Then

$$L = \frac{u + 7}{18}, \quad \theta = \frac{9u + 14}{36}, \quad A = \frac{(u + 1)^2}{9(9u + 14)}.$$

The rank-two branches meet at  $y = 2/5$ , corresponding to

$$a_c = \frac{u - 1}{45}.$$

Moreover

$$\frac{2}{5} - \frac{2/9 - A}{1 - L} = \frac{2(4u^2 - 5u - 19)}{5(u - 11)(9u + 14)} > 0 \quad (0 < u < 2),$$

so  $a = A$  always lies on the Erdős–Gallai star branch. The only possible maxima are  $a = 0$ ,  $a = A$ ,  $a = a_c$ , and the stationary point

$$a_* = \frac{13u - 14}{72}$$

when it lies in the concave branch. Direct substitution gives

$$\begin{aligned} \Psi(0) - \frac{113}{324} &= \begin{cases} -u^2/324, & 0 < u \leq 1, \\ -(2 - u)^2/324, & 1 \leq u < 2, \end{cases} \\ \Psi(A) - \frac{113}{324} &= -\frac{49u^4 + 230u^3 + 383u^2 + 352u + 200}{648(9u + 14)^2}, \\ \Psi(a_c) - \frac{113}{324} &= \frac{(u - 11)(11u - 6)}{16200} < 0 \quad (u \geq 1), \end{aligned}$$

The stationary point is feasible only when

$$\frac{62}{57} \leq u \leq \frac{-20 + 2\sqrt{5659}}{109} < \frac{6}{5}.$$

On this interval,

$$\Psi(a_*) - \frac{113}{324} = \frac{137u^2 - 364u + 196}{10368} < 0,$$

because the two roots of the numerator are  $(182 \pm 56\sqrt{2})/137$ , outside the displayed feasible interval. Thus the local recursion is impossible. Only endpoint multipliers remain, giving the two active candidates.

#### APPENDIX B. THE TRANSITION $B_{r,r-2} = B_{r,r-1}$ FOR $r \geq 5$

Continue with the notation  $d = r - 2 \geq 3$ ,  $b, \omega, x, D, \ell, L_0, L_1$  from [Section 14](#). The local recursion has an extra nonnegative penalty  $\beta/2$ , which we omit when proving a strict upper bound. For  $0 \leq a \leq A_* = L(L - \ell)/\theta$ , put  $y = (x - a)/(1 - L)$ . Then

$$\frac{d}{(d+1)^2} \leq y \leq \frac{1}{d+1}. \quad (74)$$

Indeed the upper bound follows from  $a \geq 0$ , since  $x/(1 - L)$  is increasing in  $L$  and equals  $1/(d+1)$  at  $L = L_1$ . For the lower bound, write

$$L = L_0 + z(L_1 - L_0), \quad 0 \leq z \leq 1.$$

Exact simplification gives

$$\begin{aligned} \frac{x - A_*}{1 - L} - \frac{d}{(d+1)^2} &= \frac{(338 - 239\sqrt{2})(d - \sqrt{2}/2)(1 - z)^2}{(169\sqrt{2} - 239)(d+1)^2} \\ &\quad \times \frac{1}{(dz + d(1 + \sqrt{2}) + z(\sqrt{2} + 2))(\sqrt{2}d + \sqrt{2} + 1 - z)} \geq 0. \end{aligned}$$

For  $d \geq 3$ ,

$$\frac{d}{(d+1)^2} > \frac{1}{d+1+\sqrt{2}}, \quad \frac{d}{(d+1)^2} > \frac{1}{2d},$$

so the interval in (74) lies strictly after the last adjacent  $B$ -crossing for clique size  $d+1$ , and it lies outside the  $A_{d+1}$  domain. Hence the only active branches of  $F_{r-1} = F_{d+1}$  there are  $P_d = B_{d+1,d}$  and  $R_{d+1}$ .

**B.1. The  $P_d$  branch.** Substitution gives

$$\begin{aligned} L(1 - L) + \lambda^2 P_d \left( \frac{x - a}{\lambda} \right) + \frac{\theta a}{2} - F_r(x) \\ = -(L - L_0)^2 + C_P(L)a - \frac{d^2 a^2}{2}, \end{aligned}$$

where

$$C_P(L) = d^2 x - d(1 - L) + \frac{\theta}{2}.$$

The function  $C_P$  is affine increasing. Direct simplification gives

$$C_P(L_0) = -\frac{d((2 + \sqrt{2})d^3 + (10 + 4\sqrt{2})d^2 + (12 + 2\sqrt{2})d + 4)}{4(d^2 + 4d + 2)^2} < 0,$$

and

$$\frac{C_P(L_1)}{d(L_1 - L_0)} = \frac{(2 - \sqrt{2})d + 2 + \sqrt{2}}{2d} < \sqrt{2} \quad (d \geq 3).$$

Consequently

$$C_P(L)_+ < \sqrt{2} d(L - L_0).$$

Hence

$$\max_{a \geq 0} \left( C_P(L)a - \frac{d^2 a^2}{2} \right) = \frac{C_P(L)_+^2}{2d^2} < (L - L_0)^2.$$

The branch is therefore strictly below  $F_r(x)$ .

**B.2. The  $R_{r-1}$  branch.** Write

$$\begin{aligned} L(1-L) + \lambda^2 R_{r-1} \left( \frac{x-a}{\lambda} \right) + \frac{\theta a}{2} - F_r(x) \\ = G_0(L) + C_R(L)a + Ka^2, \end{aligned}$$

where

$$K = \frac{d(d+1)^2}{2},$$

$$C_R(L) = (d+1)(d(b-L) - (1-L)) + \frac{\theta}{2}.$$

The constant term factors as

$$G_0(L) = \frac{(3-2\sqrt{2})(L-L_1)H_d(L)}{2(\sqrt{2}-1)^2(d+\sqrt{2}+2)},$$

with

$$H_d(L) = (d^2 + d(\sqrt{2}-1) - 6 - 3\sqrt{2})L + d(1-\sqrt{2}) + \sqrt{2} + 3.$$

The coefficient of  $L$  in  $H_d$  is

$$(d-3)(d+\sqrt{2}+2),$$

and

$$H_d(L_0) = \frac{4d - \sqrt{2}(d+1)}{2} > 0.$$

Thus  $H_d(L) > 0$  and  $G_0(L) < 0$  for  $L_0 < L < L_1$ .

The affine identities  $\theta(L_0) = dL_0$  and  $\theta(L_1) = (d+1)L_1$  imply  $dL < \theta < (d+1)L$  in the open interval. Together with  $0 \leq a \leq A_* = L(L-\ell)/\theta$ , this gives

$$C_R(L) + KA_* < 0.$$

For completeness, an upper bound is

$$C_R(L) + KA_* \leq (d+1) \left[ db - 1 + \frac{4-d}{2}L - \frac{d+1}{2}\ell \right],$$

The bracket is increasing in  $L$  for  $d=3$ , constant for  $d=4$ , and decreasing for  $d \geq 5$ . After multiplication by  $d+1$ , its maximum is respectively

$$\frac{2(5-4\sqrt{2})}{-3+4\sqrt{2}}, \quad \frac{5(6-5\sqrt{2})}{2(-4+5\sqrt{2})},$$

and

$$\frac{(4-3\sqrt{2})(d+1)(d+2+2\sqrt{2})}{4(\sqrt{2}-1)(d+\sqrt{2}+2)},$$

all of which are negative. Therefore

$$C_R(L)a + Ka^2 \leq a(C_R(L) + KA_*) \leq 0,$$

and the branch is strictly below  $F_r(x)$ .

APPENDIX C. THE  $r = 4$  TRANSITION  $B_{4,2} = B_{4,3}$ 

Let

$$\begin{aligned} x_0 &= \frac{4 - \sqrt{2}}{14}, & F &= \frac{3(29 - 2\sqrt{2})}{196}, \\ D = 1 - 2F &= \frac{11 + 6\sqrt{2}}{98}, & \ell = 1 - 4x_0 &= \frac{2\sqrt{2} - 1}{7}. \end{aligned}$$

The endpoint slopes are

$$m_2 = \frac{3 + \sqrt{2}}{7}, \quad m_3 = \frac{3(2 + 3\sqrt{2})}{14},$$

and the endpoint peel sizes are

$$L_2 = \frac{3 + \sqrt{2}}{14}, \quad L_3 = \frac{2 + 3\sqrt{2}}{14}.$$

Assume  $L_2 < L < L_3$ , set  $\theta = (L - D)/x_0$ ,  $\lambda = 1 - L$ , and

$$A(L) = \frac{L(L - \ell)}{\theta}.$$

The local recursion would require

$$F \leq \Phi_L(a) := L(1 - L) + \lambda^2 F_3 \left( \frac{x_0 - a}{\lambda} \right) + \frac{\theta a}{2}$$

for some  $0 \leq a \leq A(L)$ . Put  $y = (x_0 - a)/\lambda$  and write

$$L = L_2 + z(L_3 - L_2), \quad 0 < z < 1.$$

Since  $0 \leq a \leq A(L)$ ,

$$\frac{x_0 - A(L)}{1 - L} \leq y \leq \frac{x_0}{1 - L}.$$

Exact simplification gives

$$\frac{x_0 - A(L)}{1 - L} - \frac{2}{9} = \frac{(113\sqrt{2} - 144)(1 - z)^2}{(504\sqrt{2} - 1134)(z + (2 + 3\sqrt{2})/7)(z - 1 - 3\sqrt{2})} > 0,$$

and

$$\frac{1}{3} - \frac{x_0}{1 - L} = \frac{(1 - 2\sqrt{2})(z - 1)}{(3 - 6\sqrt{2})(z - 1 - 3\sqrt{2})} > 0.$$

Thus  $2/9 < y < 1/3$  throughout the strict multiplier region, so the only active branches of  $F_3$  are  $A_3$ ,  $B_{3,2}$ , and  $R_3$ . We also record the two endpoint inequalities needed below. Since  $L > L_2$ ,

$$\frac{x_0}{1 - L} > \frac{x_0}{1 - L_2} = \frac{6 - \sqrt{2}}{17} > \frac{1}{4}.$$

Furthermore the function

$$y_A(L) = \frac{x_0 - A(L)}{1 - L}$$

is decreasing on  $[L_2, L_3]$ . Indeed, after the substitution  $L = L_2 + z(L_3 - L_2)$ , direct differentiation gives

$$\frac{dy_A}{dz} = \frac{(596 - 401\sqrt{2})(z - 1)(z + 2\sqrt{2} + 3)}{(-28 + 7\sqrt{2})^2 \left( z + \frac{2+3\sqrt{2}}{7} \right)^2 (z - 3\sqrt{2} - 1)^2} < 0$$

for  $0 < z < 1$ ; here  $596 > 401\sqrt{2}$ . Hence

$$y_A(L) \leq y_A(L_2) = \frac{5 + 2\sqrt{2}}{34} < \frac{1}{4}.$$

Consequently, whenever the  $R_3$  branch is nonempty its feasible endpoints are  $a = 0$  and the boundary with  $B_{3,2}$ , while the  $A_3$  branch has endpoints at the boundary with  $B_{3,2}$  and at  $a = A(L)$ .

**C.1. The  $B_{3,2}$  branch.** The difference is

$$H_B(L, a) = -(L - L_2)^2 + C(L)a - 2a^2,$$

where

$$C(L) = 4x_0 - 2(1 - L) + \frac{\theta}{2}.$$

The affine function  $2\sqrt{2}(L - L_2) - C(L)$  has endpoint values

$$\frac{3 + \sqrt{2}}{14}, \quad \frac{26 - 17\sqrt{2}}{28},$$

which are positive. Hence

$$C(L) < 2\sqrt{2}(L - L_2).$$

If  $C(L) \leq 0$  the result is immediate; otherwise

$$\max_{a \geq 0} (C(L)a - 2a^2) = \frac{C(L)^2}{8} < (L - L_2)^2.$$

Hence  $H_B < 0$ .

**C.2. The  $R_3$  branch.** The difference  $H_R(L, a)$  is convex in  $a$ , so its maximum on the branch occurs at  $a = 0$  or at the boundary with  $B_{3,2}$ . The latter is already handled. At  $a = 0$ ,

$$H_R(L, 0) = -\frac{1}{2} \left( \frac{22 - 9\sqrt{2}}{14} - L \right) (L_3 - L) < 0.$$

**C.3. The  $A_3$  branch.** On  $2/9 \leq y \leq 1/4$  use the chord

$$A_3(y) \leq \frac{5}{36} + \frac{17}{18}y,$$

since

$$\frac{5}{36} + \frac{17}{18}y - A_3(y) = -\frac{(4y - 1)(9y - 2)}{18} \geq 0.$$

The resulting upper bound is affine in  $a$ , so it suffices to check the boundary with  $B_{3,2}$  and  $a = A(L)$ . The first is already handled. Put

$$z = \frac{L - L_2}{L_3 - L_2} \in (0, 1).$$

At  $a = A(L)$  the difference is

$$-\frac{(173 + 81\sqrt{2})Q(z)}{49392(2\sqrt{2} - 1) \left( z + \frac{2+3\sqrt{2}}{7} \right)},$$

where

$$Q(z) = q_0(1 - z)^3 + 3q_1z(1 - z)^2 + 3q_2z^2(1 - z) + q_3z^3$$

and

$$\begin{aligned} q_0 &= \frac{4(177 - 122\sqrt{2})}{7}, & q_1 &= \frac{1101 - 724\sqrt{2}}{21}, \\ q_2 &= \frac{8(131\sqrt{2} - 176)}{21}, & q_3 &= \frac{6(214\sqrt{2} - 299)}{7}. \end{aligned}$$

All four Bernstein coefficients are positive: each assertion follows by squaring the corresponding positive rational comparison with  $\sqrt{2}$  (for example,  $177 > 122\sqrt{2}$  and  $214\sqrt{2} > 299$ ). Thus  $Q(z) > 0$  and the difference is negative. This completes the final low-rank transition calculation.